# LEYUAN ANIMAL HUSBANDRY WEIXIAN CO., LTD. NO. 4 PASTURE WASTES RECYCLING PROJECT

Document Prepared by Climate Bridge (Shanghai) Ltd.

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| **Project Title** | Leyuan ANIMAL Husbandry Weixian Co., Ltd. NO. 4 Pasture Wastes Recycling Project |
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# 1 PROJECT DETAILS

## 1.1 Summary Description of the Project

Leyuan ANIMAL Husbandry Weixian Co., Ltd. NO. 4 Pasture Wastes Recycling Project (hereinafter referred to as the Project) locates in Xiwang Village, Houguan Town, Weixian District, Xingtai City, Hebei Province, P.R China. The project is owned and implemented by Hebei Zhonghai Huaneng Energy Co., Ltd

The project has built a centralized anaerobic animal manure treatment system which collects manure waste (dairy cattle) from Leyuan ANIMAL Husbandry Weixian Co., Ltd. NO. 4 Pasture in Weixian district. The project uses animal manure waste as ferment material to produce biogas in the Completely Stirred Tank Reactor (CSTR) type anaerobic digesters. The recovered biogas is used for electricity generation at the project site. The electricity generated by the project is delivered to North China Power Grid (NCPG). The residual waste from the CSTR digesters is used to produce organic fertilizer at the project site.

According to the feasibility study report, the intended installed capacity of the electricity generator is 3.83MW (1.165MW * 2 + 1.5MW *1). The project is divided into two phases. In phase Ⅰ, the installed capacity will be 2.33MW (1.165MW * 2) and in phase Ⅱ, the installed capacity will be 1.5MW. The construction of phase Ⅱ will depend on the circumstances in the near future.

Prior to the implementation of the project, the animal manure waste was left to decay in anaerobic manure management system (lagoon) at the livestock farms and methane is emitted to the atmosphere directly without any methane recovery and destruction facility. Equivalent amount of electricity replaced by the Project are generated by the fossil fuel fired power plants of NCPG.

The project is expected to avoid GHG emission of methane through recovery and destruction of biogas. In addition, the electricity generated from the biogas is delivered to NCPG to replace equivalent electricity generated by the fossil fuel fired power plants connected to the grid. The project is expected to achieve an annual emission reduction of 57,220 $tCO_2e$ and a total emission reduction of 400,540 $tCO_2e$ during the first 7-year crediting period. During this monitoring period from 21-Mar-2021 to 30-Sep-2022, the actual operating capacity of the project is 2.33MW (2 sets of 1.165MW CMM generators have been operated to generate electricity), the achieved emission reductions are 47,385 $tCO_2e$.

## 1.2 Sectoral Scope and Project Type

Sectoral Scope 1: Energy industries (renewable - / non-renewable sources)

Sectoral Scope 13: Waste handling and disposal.

The project is not a grouped project.

## 1.3 Project Eligibility

The scope of the VCS Program includes:

1) The seven Kyoto Protocol greenhouse gases: The project is expected to avoid two greenhouse gases: 1) Methane ($CH_4$) emissions from the anaerobic animal manure management system in the baseline scenario, which will be captured and destroyed in the project scenario; 2) $CO_2$ emissions from the production of the equivalent amount of electricity replaced by the Project that would otherwise be generated by the fossil fuel fired power plants of NCPG. Thus, the project is applicable to this scope.

2) Ozone-depleting substances (ODS): Not Applicable.

3) Project activities supported by a methodology approved under the VCS Program through the methodology development and review process: Not Applicable.

4) Project activities supported by a methodology approved under an approved GHG program, unless explicitly excluded (see the Verra website for exclusions): The applied methodology AMS-III.D (Version 21.0) and AMS-I. D (Version 18.0) of the project are methodologies approved under CDM Program, which is a VCS approved GHG program. Both the two methodologies are not explicitly excluded.

5) Jurisdictional REDD+ programs and nested REDD+ projects as set out in the VCS Program document Jurisdictional and Nested REDD+ (JNR) Requirements: Not Applicable.

Furthermore, the project does not belong to the project activities excluded in Table 1 of VCS Standard 4.4.

Thus, the project is eligible under the scope of VCS program.

## 1.4 Project Design

The project has been designed to be a single installation of an activity, not a grouped project.

### Eligibility Criteria

The project is not a grouped project. There is no additional information needed to be provided.

## 1.5 Project Proponent

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|---|---|
| Organization name | Hebei Zhonghai Huaneng Energy Co., Ltd |

| | |
|---|---|
| Contact person | Ran Qiyan |
| Title | Manager |
| Address | Xiwang Village, Houguan Town, Weixian District, Xingtai City, Hebei Province, P.R China |
| Telephone | +86-021 23019950 |
| Email | 3542346576@qq.com |

## 1.6 Other Entities Involved in the Project

| | |
|---|---|
| Organization name | Climate Bridge (Shanghai) Ltd. |
| Role in the project | consultant |
| Contact person | Gao Zhiwen |
| Title | General Manager |
| Address | Block B, Level 24, Jiangong Mansion, 33 Fushan Road, Pudong New Area, Shanghai, China 200120 |
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| Email | gao.zhiwen@climatebridge.com |

## 1.7 Ownership

The project proponent of the project is Hebei Zhonghai Huaneng Energy Co., Ltd. The project proponent has obtained the approval from Xingtai Municipal Administrative Examination and Approval Bureau for the construction and operation of the Project. The Environmental Impact Assessment (EIA) of the Project prepared by Hebei Shihuan Environmental Assessment Service Co., Ltd. shows that the Project is constructed and operated by the project proponent, and the EIA form has been approved by Weixian Branch of Xingtai Ecological and Environmental Bureau.

Both the project approval and the EIA approval are issued by competent authority, and they demonstrate that the project proponent has been granted the project ownership in compliance with relevant laws and regulations in China.

In addition, the equipment purchasing contracts and the construction contract signed between the project proponent and suppliers/contractors demonstrate the ownership of the project proponent over the project.

## 1.8 Project Start Date

21-Mar-2021 (the start date of operation)[1].

## 1.9 Project Crediting Period

This project adopts a 7-year renewable crediting period. The first crediting period is from 21-Mar-2021 to 20-Mar-2028 (both days included).

## 1.10 Project Scale and Estimated GHG Emission Reductions or Removals

| Project Scale | |
|---|---|
| Project | √ |
| Large project | |

| Year | Estimated GHG emission reductions or removals ($tCO_2e$) |
|---|---|
| 21-Mar-2021~31-Dec-2021 | 44,835 |
| 01-Jan-2022~31-Dec-2022 | 57,220 |
| 01-Jan-2023~31-Dec-2023 | 57,220 |
| 01-Jan-2024~31-Dec-2024 | 57,220 |
| 01-Jan-2025~31-Dec-2025 | 57,220 |
| 01-Jan-2026~31-Dec-2026 | 57,220 |
| 01-Jan-2027~31-Dec-2027 | 57,220 |
| 01-Jan-2028~ 20-Mar-2028 | 12,385 |
| Total estimated ERs | 400,540 |

[1] The start date of the project is from daily operation log.

| | |
|---|---|
| Total number of crediting years | 7 |
| Average annual ERs | 57,220 |

## 1.11 Description of the Project Activity

The project involves construction of a centralized animal manure management system which collects manure waste (dairy cattle) from Leyuan ANIMAL Husbandry Weixian Co., Ltd. NO. 4 Pasture in Weixian district. The project uses animal manure waste as ferment material to produce biogas in the Completely Stirred Tank Reactor (CSTR) type anaerobic digesters. The anaerobic digesters are expected to produce 13,760,000 $m^3$ biogas per year. The recovered biogas will be used for electricity generation at the biogas power plant. Waste heat recovery boiler will also be installed at the biogas power plant to recover and supply waste heat to the auxiliary equipment of the project. The emission reductions from waste heat recovery will not be accounted for. The installed capacity of the biogas power plant will be 3.83MW (1.165MW * 2 + 1.5MW *1). The biogas power plant of the project is expected to supply 27,000 MWh[2] of electricity to the North China Power Grid annually. The project activity is expected to have a 20-year operational lifetime.

The project is expected to avoid GHG emission of methane through recovery and destruction of biogas. In addition, the electricity generated from the biogas will be delivered to NCPG to replace equivalent electricity generated by the fossil fuel fired power plants connected to the grid.

The key system involved in the project are as follows:

Anaerobic animal manure management system:

The Completely Stirred Tank Reactor (CSTR) type anaerobic digesters are to be applied in the project activity. Three CSTR digesters with effective volume of 6,468.33$m^3$ each and 19,405 $m^3$ in total will be installed. The project is expected to produce 13,760,000 $m^3$ biogas per year(Phase Ⅰ: 8,760,000 $m^3$; phase Ⅱ:5,000,000 $m^3$). The biogas produced from this procedure will be transported to biogas pre-treatment system.

Biogas pre-treatment system

Three integrated biogas storage cabinets with effective volume of 2387.42$m^3$ each and 7162.26$m^3$ in total will be installed to store biogas before electricity generation, combustion or flaring, the biogas will be stored and pre-treated to remove impurities and moisture etc., to prevent the corrosion of the project facility. In addition, biogas should be continuously in a stable condition before it flows into gas engine or flare. Only when emergency occurs, the enclosed flare system will be used and the biogas will be flared. The processing capacity of flare system is

[2] According to FSR

1000m³/h, the diameter is 1.7m and the height is 10.5m. The pre-treatment system, consists of biodesulfurization, dehumidification, dewatering, cooling and fine desulfurization.

Organic fertilizer production plant

The digestate produced after anaerobic fermentation is separated by solid-liquid separation, and the solid part is squeezed twice before entering the drying drum. The dried part is used as cow bedding material, and the liquid part is treated with aerobic aeration before being returned to farmland for fertilizer. The liquid part will be sent to aerobic treatment as soon as digestate is separated by solid-liquid separation. Digestate will not be stored under anaerobic conditions. In addition, the project does not involve anaerobic composting of digestate. Hence proper conditions and procedures (not resulting in methane emissions) for digestate is ensured.

Biogas power generation system

The biogas electricity generation plant will be installed at the project site. The total installed capacity will be 3.83MW (1.165MW * 2 + 1.5MW *1). The biogas power plant of the project is expected to supply 27,000 MWh of electricity to the North China Power Grid annually. Waste heat recovery boiler will also be installed at the biogas power plant to recover and supply waste heat to the auxiliary equipment of the project.

Table 1: Facilities Specifications[3]

| Type | Combustion Engine | Generator |
|---|---|---|
| Model | 12V4000L32FB | HVSI804R2 |
| Quantity | 2 | 2 |
| Rated Power | 1.165MW | 1.734MW |
| Rated speed (r/min) | 1,500 | / |
| Life time (year) | 20 | 20 |

## 1.12 Project Location

The Project locates in Xiwang Village, Houguan Town, Weixian District, Xingtai City, Hebei Province, P.R China. The geographical coordinates for the project site are east longitude 115°26′20.24″ and north latitude 37°07′58.57″.

The geographic location of the project is shown in Figure 1-1

[3] The running capacity of the project is 2.33MW (2 sets of 1.165MW CMM generators have been operated to generate electricity), while the intended installed capacity is 3.83MW (1.165MW * 2 + 1.5MW *1). The installation of the other 1.5MW will depend on the circumstances. The information is from the technical agreement of equipment.

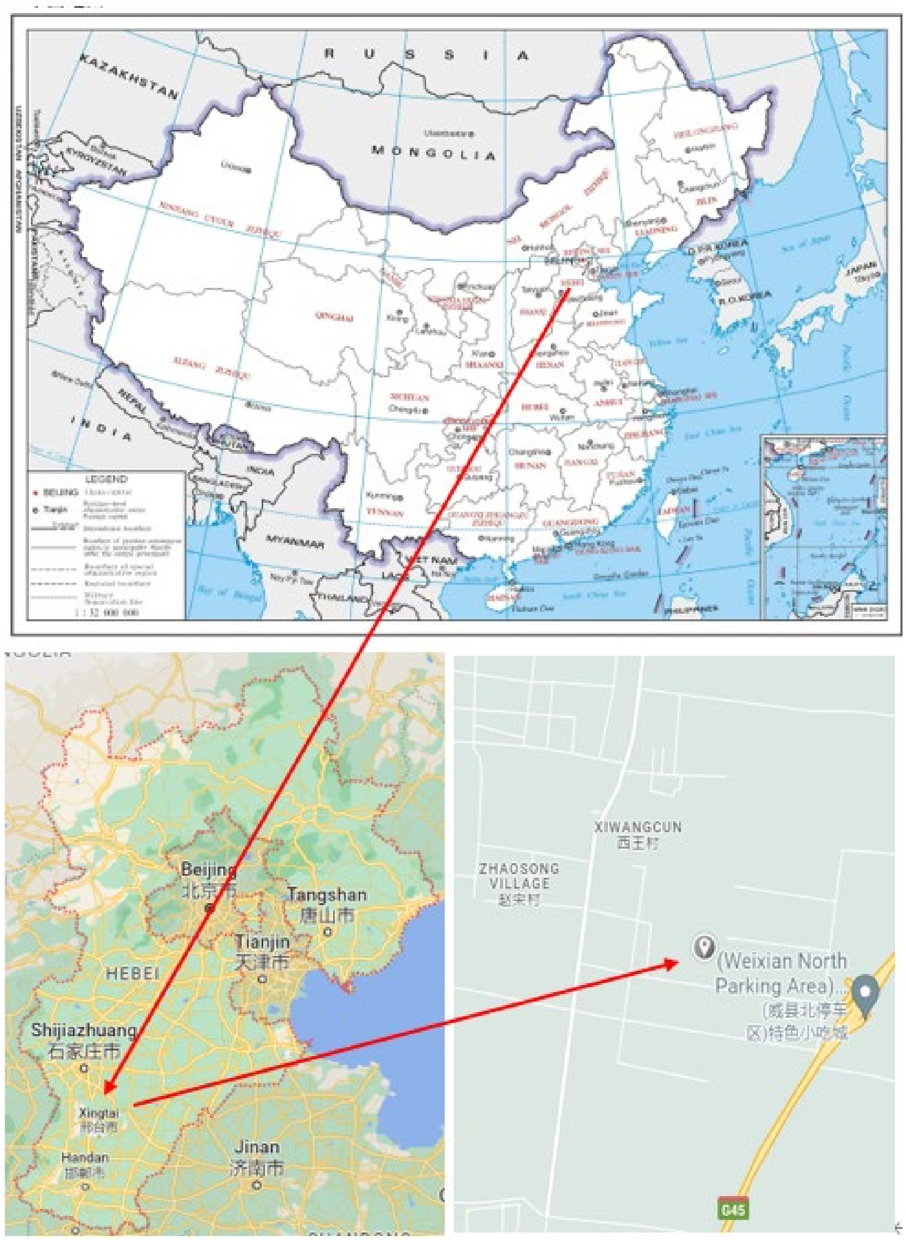


**Figure 1-1 The geographic location of the project**

## 1.13 Conditions Prior to Project Initiation

The condition existing prior to project initiation is the same as the baseline scenario of the project, please refer to Section 3.4(Baseline Scenario).

## 1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

The project complies with all Chinese relevant laws and regulations. Mainly include:

1. Environmental Protection Law of People's Republic of China[4];
2. Law of the People's Republic of China on the Prevention and Control of Solid Waste Pollution[5];
3. Regulations on prevention and control of pollution from large scale livestock and poultry breeding[6]
4. Renewable Energy Law of the People's Republic of China[7];
5. Catalogue for the Guidance of Industrial Structure Adjustment (2019 revision)[8];

The project has obtained the project approval and EIA approval from local government authorities: Xingtai Municipal Administrative Examination and Approval Bureau and Weixian Branch of Xingtai Ecological and Environmental Bureau. The two approvals well demonstrate that local government permits the construction of the project. Consequently, the project is compliance with laws, status and other regulatory frameworks.

## 1.15 Participation under Other GHG Programs

### 1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project has neither been registered, nor is seeking registration under any other GHG programs. The project is seeking registration only under VCS program.

### 1.15.2 Projects Rejected by Other GHG Programs

The project has never been seeking registration under any other GHG program; hence the project has never been rejected by any other GHG programs.

## 1.16 Other Forms of Credit

### 1.16.1 Emissions Trading Programs and Other Binding Limits

China has a national emissions trading scheme only cover the high-emission industries, such as thermal power generation, petrochemical, chemical, building materials, iron and steel, nonferrous, paper, aviation and other key emission industries that emitted at least 26,000 tons

[4] http://www.china-npa.org/uploads/1/file/public/201803/20180327160313_j8uwkhdxgn.pdf

[5] https://www.mee.gov.cn/ywgz/fgbz/fl/202004/t20200430_777580.shtml

[6] http://www.gov.cn/flfg/2013-11/26/content_2535095.htm

[7] http://www.nea.gov.cn/2017-11/02/c_136722869.htm

[8] http://www.gov.cn/xinwen/2019-11/06/content_5449193.htm

of $CO_2$e/year, not including renewable project[9].

Also, the project has not been registered as a CCER (Chinese Certified Emission Reductions) project in China, thus it is not eligible for emission reductions transaction under the China's ETS. Therefore, the project does not reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading. The net GHG emission reductions generated during this monitoring period have not been used for compliance under such programs or mechanisms.

Furthermore, as a voluntary emission reduction project, the GHG emission reductions and removals generated by the project will not be enforced to be used for compliance in China. The credits in this monitoring period have not been counted and will not be counted under emission trading programs and other binding limits.

### 1.16.2 Other Forms of Environmental Credit

Based on the application requirements of other GHG programs, such as CDM, VCS, GS, GCC, renewable energy certificates, etc, and the suspension of CCER, the project can only apply for VCS and GCC programs, while the project proponent chooses to apply for VCS programs. Thus, the project has not sought or received another form of GHG-related environmental credit. None of the credits will be claimed in the same period.

### 1.16.3 Supply Chain (Scope 3) Emissions

According to the Verra document,[10] in this joint PD&MR, this section is not required.

## 1.17 Sustainable Development Contributions

### 1.17.1 Sustainable Development Contributions Activity Description

The project uses animal manure waste as ferment material to produce biogas for electricity generation. By destroying biogas that would have been vented into the atmosphere directly in the absence of the project activity, and by displacing part of the electricity that would have been supplied by the grid, the project achieves significant GHG emission reductions. The project also provides long-term job opportunities to local people and contributes to the economic prosperity.

The Project will contribute to sustainable development in the following ways:

- Increase electricity production from renewable sources, which will certainly reduce the consumption of fossil fuel in North China Power Grid and further help to achieve the national action to promote renewable energy development. Thus, the project achieved SDG 7

[9] http://www.mee.gov.cn/xxgk2018/xxgk/xxgk05/202103/t20210330_826728.html

[10] https://verra.org/verra-releases-updates-to-verified-carbon-standard-program/

Affordable and Clean Energy[11].

- Reduce greenhouse gas emissions. The project will recover and destroy biogas that consists mainly of greenhouse gas methane and would otherwise be released directly into the atmosphere, effectively reducing greenhouse gas emissions, which contributes to China's commitment to peak carbon dioxide emissions before 2030. Thus, the project achieved SDG 13 Climate Action[12].

- The project will provide job opportunities and increase tax revenue, which will have a positive effect on the local economy. Thus, the project achieved SDG 8 Decent Work and Economic Growth[13].

The project contributes to achieving Chinese sustainable development priorities is described as below:

| SDG | Indicators | Chinese Sustainable Development Progress | Project activity contribution |
|---|---|---|---|
|  | SDG 7 "Ensure access to affordable, reliable, sustainable and modern energy for all" | By the end of 2015, China had fully solved the problem of providing electricity to all people without electricity, and reached the goal of the Sustainable Development Agenda 15 years ahead of schedule. From 2015 to 2020, the share of non-fossil energy increased from 14.5 % to 19.6%, while the share of raw coal decreased from 72.2 %to 67.6%%. | The project is to capture and utilize biogas for power generation and accordingly substitute equivalent electricity from thermal power. This contributes to achieve China's stated sustainable development priorities "By 2030, the proportion of non-fossil energy consumption will reach about 25%". |
|  | SDG 8 "Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all". | China has deeply implemented the innovation-driven development strategy. Small, medium and micro enterprises have developed rapidly. Adhering to the policy of giving priority to | The project activity can provide employment opportunities for local villagers. This contributes to one of China's actions for promoting sustainable developing: "by 2030, achieve full and productive employment |

11 https://sdgs.un.org/goals/goal7

12 https://sdgs.un.org/goals/goal13

13 https://sdgs.un.org/goals/goal8

| | | | |
|---|---|---|---|
| | | employment, the unemployment rate has remained at a low level. By coordinating epidemic prevention and control with economic and social development, China has become the only major economy to achieve positive growth in 2020, and has made positive contributions to global economic recovery. | and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value". |
|  | SDG 13 "Take urgent action to combat climate change and its impacts". | In 2020, China's energy consumption per unit of GDP was reduced by 24.4% compared with 2012; carbon dioxide emissions per unit of GDP was reduced by 18.8% compared with 2015 and 48.4% compared with 2005, all of which have already fulfilled China's commitment to the international community in 2020 ahead of schedule. | The project is to capture and utilize biogas for power generation to avoid biogas emissions and $CO_2$ emissions from the production of the equivalent amount of electricity replaced by the Project that would otherwise have been purchased from the NCPG. This contributes to achieve China's stated sustainable development priorities " China's carbon dioxide emissions would strive to peak by 2030 and strive to achieve carbon neutrality by 2060 ". |

## 1.17.2 Sustainable Development Contributions Activity Monitoring

The project, located in Xiwang Village, Houguan Town, Weixian District, Xingtai City, Hebei Province, P.R China, utilizes biogas to generate power. By utilizing animal manure waste as ferment material to produce biogas and by displacing part of the electricity that would have been supplied by the grid, the project achieves significant GHG emission reductions. The project is owned and implemented by Hebei Zhonghai Huaneng Energy Co., Ltd. The project also provides long-term job opportunities for local people and contributes to the economic prosperity.

**Table 1-1: Sustainable Development Contributions**

| Row number | SDG Target | SDG Indicator | Net Impact on SDG Indicator | Current Project Contributions | Contributions Over Project Lifetime |
|---|---|---|---|---|---|
| 1) | 7.2 | 7.2.1: Renewable energy share in the total final energy consumption | Implemented activities to increase | 17,296.840 MWh of electricity from renewable sources have been supplied to the power grid during the first monitoring period (21-Mar-2021~30-Sep-2022), replacing fossil fuel powered electricity. | From 21-Mar-2021 to 30-Sep-2022, 17,296.840 MWh of electricity from renewable sources has been supplied to the power grid. |
| 2) | 8.5 | The number of the job opportunities provided by the implementation of the project | Implemented activities to increase | Provided 19 employment positions for local people. No further changes in this reporting period. | Provided 19 employment positions for local people in the reporting period. |
| 3) | 13.0 | Tonnes of greenhouse gas emissions avoided or removed | Implemented activities to increase | The project achieved a GHG emission reduction of 47,385 $tCO_2e$ during the first monitoring period(21-Mar-2021~30-Sep-2022). | From 21-Mar-2021 to 30-Sep-2022, the project has achieved a GHG emission reduction of 47,385 $tCO_2e$. |

## 1.18 Additional Information Relevant to the Project

### Leakage Management

Leakage emissions are considered as per the applicable tool:

According to AMS-III.D Methane recovery in animal manure management systems (version 21.0): Leakage is determined by following the relevant procedure in the methodological tool “Project and leakage emissions from anaerobic digesters”.

As per “Project and leakage emissions from anaerobic digesters” (version 02.0): The following sources of leakage emissions are accounted for in this tool:

(a) $CH_4$ and $N_2O$ emission from composting of digestate;

(b) $CH_4$ emissions from the anaerobic decay of digestate disposed in a SWDS or subjected to anaerobic storage, such as in a stabilization pond.

The leakage emissions associated with the anaerobic digester depend on how the digestate is managed. The leakage emissions include emissions associated with storage of digestate and emissions associated with composting of the digestate.

The digestate produced after anaerobic fermentation is separated by solid-liquid separation, and the solid part is squeezed twice before entering the drying drum. The dried part is used as cow bedding material, and the liquid part is treated with aerobic aeration before being returned to farmland for fertilizer. The liquid part will be sent to aerobic treatment as soon as digestate is separated by solid-liquid separation. Digestate will not be stored under anaerobic conditions. In addition, the project does not involve anaerobic composting of digestate. Therefore, leakage emissions of the project associated with the anaerobic digester is not accounted for.

### Commercially Sensitive Information

There is no commercially sensitive information has been excluded from the public version of the project description.

### Further Information

There is no additional relevant legislative, technical, economic, sectoral, social, environmental, geographic, site-specific and/or temporal information that may have a bearing on the eligibility of the project, the net GHG emission reductions or rem ovals, or the quantification of the project’s net GHG emission reductions or removals.

# 2 SAFEGUARDS

## 2.1 No Net Harm

The Environmental Impact Assessment (EIA) of the Project prepared by Hebei Shihuan Environmental Assessment Service Co., Ltd. has been approved by Weixian Branch of Xingtai Ecological and Environmental Bureau on 04-Dec-2019, with approval No. "Xinghuanweishen [2019] No.79". Every aspect of environmental impact has been considered in the EIA report with corresponding measures during project development. The environmental impacts, treatment, and effects arising from the Project during operation have been analysed in section 2.3 of this report below. And no net harm has been detected.

Meanwhile, the implementation of the project will improve local socio-economic development and contribute to sustainable development as described in section 1.17 of this report above.

## 2.2 Local Stakeholder Consultation

The Project owner collected comments by local stakeholders on the project activity. Survey questionnaires were distributed to relevant personnel of the pasture, local villagers and government officials by the project owner in March 2021. The survey questionnaire was designed to assess the project impacts on the local environment and social economic development. The structure of the survey respondents is listed in Table 2-1 below.

**Table 2-1 Structure of stakeholder survey**

| Item \ Distribution | | Quantity | Percentage |
|---|---|---|---|
| Number of stakeholders surveyed | Male | 15 | 50% |
| | Female | 15 | 50% |
| Age | <25 | 6 | 20% |
| | 25-55 | 18 | 60% |
| | >55 | 6 | 20% |
| Education | Junior high school or below | 6 | 20% |

| | | | |
|---|---|---|---|
| | Senior high school | 16 | 53% |
| | College or above | 8 | 27% |
| Occupation | Worker | 14 | 47% |
| | Peasant | 9 | 30% |
| | Management personnel | 4 | 13% |
| | Civil servant | 2 | 7% |
| | Unspecified | 1 | 3% |

Thirty questionnaires were distributed to local stakeholders, and all questionnaires have been recollected. Comments from these questionnaires are summarized in Table 2-2 below:

**Table 2-2 Summary of stakeholders' comments**

| No. | Questions | Attitude or Opinion | Amount | Percentage |
|---|---|---|---|---|
| 1 | Do you know about the project activity? | Very much | 26 | 87% |
| | | Heard of | 1 | 3% |
| | | Don't know | 3 | 10% |
| 2 | Do you think the project will improve the current situation of livestock farms? | Yes | 27 | 90% |
| | | No | 0 | 0 |
| | | Don't know | 3 | 10% |
| 3 | Do you think the project will improve the local employment situation? | Yes | 25 | 83% |
| | | No | 0 | 0 |
| | | Don't know | 5 | 17% |
| 4 | | Yes | 20 | 67% |

| | | | | |
|---|---|---|---|---|
| | Do you think the project will improve the local social community? | No | 0 | 0 |
| | | Don't know | 10 | 33% |
| 5 | Do you think the project will promote local economic development? | Positive impact | 20 | 67% |
| | | No impact | 10 | 33% |
| | | Negative impact | 0 | 0% |
| 6 | What is the most probable environmental impact do you think the project will cause after the construction finish? (Multiple choice) | None | 24 | 80% |
| | | Air pollution | 0 | 0% |
| | | Water pollution | 0 | 0 |
| | | Noise pollution | 0 | 0% |
| | | Harm to indigenous animals and plants | 0 | 0% |
| | | Don't know | 6 | 20% |
| 7 | What is your attitude towards the project activity? | Support | 27 | 90% |
| | | Against | 0 | 0 |
| | | Indifferent | 3 | 10% |

In general, local stakeholders are supportive of the project construction. The survey shows that a majority of local stakeholders think the Project will help improve the life of local people and promote local economic development without much adverse environmental impact. The survey shows that almost all of the stakeholders are supportive to the proposed project, believing that the Project will provide more employment opportunities, help local economic development, and increase local power supply. Therefore, the implementation of the Project is regarded as beneficial by most of the local stakeholders.

The mechanism for ongoing communication with local stakeholders:

Communications with Local stakeholders will be carried out at periodic intervals. The project proponent uses the following methods to communicate with stakeholders: 1) Survey questionnaires are distributed to relevant personnel of the pasture, local residents and government officials. For personnel of the pasture, the project owner goes to the pasture to collect the opinion of them. For local villagers, the project owner contacts the responsible person of local community who informs villagers to participate in the survey, thus, the questionnaires are distributed. For government officials, the project owner goes to the office buildings of local government to distribute questionnaires to collect their opinions. 2) The project proponent has put their phone number on a bulletin board at the gate of the project site, stakeholders can get in touch with them as long as they have any suggestion or complains. The feedback of suggestions and complains received will be put on the board for at least two weeks. During this monitoring period, there were 30 questionnaires distributed on 22-Jun-2022 and according to the result there were no negative comments received. Also, the project proponent received no comments for the project by phone call.

Key implementation schedules or changes of the project will be communicated to the local authority, who will inform the neighbourhood committee and the local residents. The comments and suggestions from residents will be collected by the local authority meanwhile. And the local government agencies and competent authorities will conduct spot checks on the implementation of the project from time to time and give suggestions on the involved rectification problems. In line with VCS requirements, all the processes have been implemented to receive comments from local stakeholders as well as communicate with them at periodic intervals.

## 2.3 Environmental Impact

The Environmental Impact Assessment (EIA) of the Project prepared by Hebei Shihuan Environmental Assessment Service Co., Ltd. has been approved by Weixian Branch of Xingtai Ecological and Environmental Bureau on 04-Dec-2019, with approval No. “Xinghuanweishen [2019] No.79”. The environmental impacts of the project during construction and operation are summarized as follows.

**1.Construction Phase**

1.1 Air pollution

Construction dust and road dust during the construction has a certain effect to the surrounding areas of construction site, these adverse effects are accidental, temporary and partial, as long as to take effective measures and strengthen management, the scope of its influence is generally limited to the surrounding area of the construction site and will disappear along with the end of construction.

1.2 Wastewater

Wastewater during the construction period mainly includes construction wastewater and domestic sewage. The following measures are taken to prevent and control the pollution of construction wastewater: construct temporary diversion ditches on the construction site; set up a sedimentation tank, and reuse the washing water for construction machinery as much as possible after simple treatment of equipment and vehicles.

Domestic sewage can be collected and processed by setting up mobile toilets, and regularly handed over to the sanitation department to remove garbage and excrement, which can effectively prevent the sewage generated by construction workers from polluting the water environment.

1.3 Noise

The noise generated by the project construction has a slight impact on the surrounding sensitive points, and its pollution impact is localized and short-term. After adopting a reasonable construction organization method, its impact on the surrounding area can be reduced to acceptable range.

1.4 Solid waste

During the construction period, a certain amount of construction waste and domestic waste from construction workers will be generated, of which part of the construction waste will be recycled. The unusable waste will be handed over to the sanitation department for disposal. After the above measures are taken, the solid waste of the project will not cause pollution to the surrounding environment

**2.Operation Phase**

2.1 Air pollution

The air pollution during the operation mainly includes the exhaust gas of the biogas generator and the flare, and dust from organic fertilizer plant. The exhaust gas is to be treated through 2 stage purification equipment before being discharged through exhaust cylinders with 15m high. The dust from organic fertilizer plant is to be filtered by dust collector and reused in the organic fertilizer plant. The $NO_x$, sulfur dioxide, $H_2S$, $NH_3$ and particulate matter in all the exhaust gas by the project meets the Table 2 standard limit of the "Odorous Pollutant Emission Standard" (GB14554-93).

2.2 Wastewater

The wastewater during operation is mainly domestic sewage and biogas slurry from anaerobic digesters. The domestic sewage of the project is pretreated in a septic tank and reused in anaerobic digesters. Most biogas slurry will also be recycled in anaerobic digesters, surplus biogas slurry will be used as material for organic fertilizer production. No wastewater will be discharged to the environment.

2.3 Noise

The high noise equipment of the project mainly consists of various fans and pumps. After sound insulation, vibration reduction and noise elimination, the noise at the plant boundary can meet the requirements of Class II standard of the "Emission Standard for Industrial Enterprises Noise at Boundary" (GB12348-2008) after greening and distance attenuation.

2.4 Solid waste

The solid wastes generated in the operation process of this project mainly include biological sludge of desulfurization system, sludge of biogas purification system, digestate, laboratory waste reagents and waste liquid, and domestic waste of employees.

1) Biological sludge of desulfurization tower biosludge

After pressure filtration, it is temporarily stored in the desulfurization tank and regularly sold to the sulfur manufacturer.

2）Sludge of biogas purification system

After dewatering by the dehydrator in the factory area, it will be collected and cleaned and disposed of by the local sanitation department.

3)Digestate

The digestate produced after anaerobic fermentation is separated by solid-liquid separation, and the solid part is squeezed twice before entering the drying drum. The dried part is used as cow bedding material, and the liquid part is treated with aerobic aeration before being returned to farmland for fertilizer. The liquid part will be sent to aerobic treatment as soon as digestate is separated by solid-liquid separation. Digestate will not be stored under anaerobic conditions. In addition, the project does not involve anaerobic composting of digestate. Hence proper conditions and procedures (not resulting in methane emissions) for digestate is ensured.

4)Laboratory waste reagents and waste liquid

Laboratory waste reagents and waste liquid will be properly collected and stored in the hazardous waste temporary deposit, and then be submitted to the relevant qualified entities for centralized treatment regularly.

5）Domestic waste of employees

It will be collected, cleaned and disposed of by the local sanitation department.

In conclusion, the environmental impact during the project construction will be temporary and not significant. The environmental impact during the project operation will be minor. The project activity can reduce greenhouse gas emissions and environmental pollution caused by methane release and coal-fired power generation. The Project owner takes appropriate measures to minimize adverse environmental impacts.

## 2.4 Public Comments

This project has been listed for public comments on the verra website during the period from 19-Dec-2022 to 18-Jan-2023 and no comments were received.

## 2.5 AFOLU-Specific Safeguards

The project is a non-AFOLU project. For non-AFOLU projects, this section is not required.

# 3 APPLICATION OF METHODOLOGY

## 3.1 Title and Reference of Methodology

The methodologies applied to the project are small scale CDM methodologies:

AMS-III.D Methane recovery in animal manure management systems (version 21.0);

AMS-I. D Grid connected renewable electricity generation (version 18.0).

This methodology also refers to the latest approved version of the following tools:

Project and leakage emissions from anaerobic digesters (version 02.0);

Tool to calculate the emission factor for an electricity system (version 07.0);

Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation" (version 03.0)

For more detail about the methodology and tools, please refer to the following link:

https://cdm.unfccc.int/methodologies/DB/H9DVSB24O7GEZQYLYNWUX23YS6G4RC

https://cdm.unfccc.int/methodologies/DB/W3TINZ7KKWCK7L8WTXFQQOFQQH4SBK

## 3.2 Applicability of Methodology

The project satisfies all the applicability criteria of the methodology AMS-III.D (Version 21.0) and AMS-I. D (version 18.0), of which the detailed description is listed in Table 3-1 below:

Table 3-1 Applicability of AMS-III.D

| No. | Applicability conditions of the methodology | The project |
|---|---|---|
| 1 | This methodology covers project activities involving the replacement or modification of anaerobic animal manure management systems in livestock farms to achieve methane recovery | Applicable.<br>The project is a centralized plant treats animal manure waste collected from |

| | | |
|---|---|---|
| | and destruction by flaring/combustion or gainful use of the recovered methane. It also covers treatment of manure collected from several farms in a centralized plant | Leyuan ANIMAL Husbandry Weixian Co., Ltd. NO. 4 Pasture. The project replaces existing anaerobic animal manure management systems (open lagoon) in livestock farms to achieve methane recovery and destruction by flaring/combustion or gainful use of the recovered methane. |
| 2 | This methodology is only applicable under the following conditions:<br><br>a) The livestock population in the farm is managed under confined conditions | Applicable.<br><br>All the livestock population in the farms within the project boundary is managed under confined conditions |
| | b) Manure or the streams obtained after treatment are not discharged into natural water resources (e.g., river or estuaries), otherwise “AMS-III.H Methane recovery in wastewater treatment" shall be applied | Applicable.<br><br>As per the FSR, manure or the streams obtained after treatment are not discharged into natural water resources (e.g., river or estuaries) |
| | c) The annual average temperature of baseline site where anaerobic manure treatment facility is located is higher than 5°C | Applicable.<br><br>The annual average temperature of baseline site where anaerobic manure treatment facility is located is higher than 5°C. The annual average temperature at the project site is 13.1℃[14]. |
| | d) In the baseline scenario the retention time of manure waste in the anaerobic treatment system is greater than one month, and if anaerobic lagoons are used in the baseline, their depths are at least 1 m | Applicable.<br><br>In the baseline scenario the retention time of manure waste in the anaerobic lagoons is greater than one month, and their depths are at least 1 m. |
| | e) No methane recovery and destruction by flaring or combustion for gainful use takes place in the baseline scenario | Applicable.<br><br>No methane recovery and destruction by flaring or combustion for gainful use takes place in the baseline scenario. |
| 3 | The project activity shall satisfy the following conditions: | Applicable. |

[14]According to FSR

| | | |
|---|---|---|
| | (a) The residual waste from the animal manure management system shall be handled aerobically, otherwise the related emissions shall be taken into account as per relevant procedures of "AMS-III.AO Methane recovery through controlled anaerobic digestion". In the case of soil application, proper conditions and procedures (not resulting in methane emissions) must be ensured; | The residual waste from the animal manure management system of the project will be used to produce organic fertilizer, which is handled aerobically and will not result in methane emissions. |
| | (b) Technical measures shall be used (including a flare for exigencies) to ensure that all biogas produced by the digester is used or flared; | Applicable.<br>Biogas produced by the project are used directly to produce electricity, the emergency flare ensure biogas would be destroyed while exigencies happened |
| | (c) The storage time of the manure after removal from the animal barns, including transportation, should not exceed 45 days before being fed into the anaerobic digester. If the project proponent can demonstrate that the dry matter content of the manure when removed from the animal barns is larger than 20%, this time constraint will not apply. | Applicable.<br>The storage time of the manure after removal from the animal barns, including transportation will not exceed 45 days before being fed into the anaerobic digester. |
| 4 | Projects that recover methane from landfills shall use "AMS-III.G Landfill methane recovery" and projects for wastewater treatment shall use AMS-III.H. Projects for composting of animal manure shall use "AMS-III.F Avoidance of methane emissions through composting". Project activities involving co-digestion of animal manure and other organic matters shall use the methodology "AMS-III.AO Methane recovery through controlled anaerobic digestion". | Irrelevant.<br>The project does not involve landfill methane recovery, wastewater treatment, composting animal manure, or co-digestion of animal manure and other organic matters. |
| 5 | Utilization of the recovered biogas in one of the options detailed in AMS-III.H is also eligible under this methodology. The respective procedures in AMS-III.H shall be followed in this regard. If the recovered biogas is used to power auxiliary equipment of the project activity, it should be | Applicable.<br>Part of the recovered biogas is used to produce and deliver electricity to the power grid, therefore AMS.I. D is applied.<br>Part of the biogas is used to power auxiliary equipment of the project activity, |

| | | |
|---|---|---|
| | taken into account accordingly, using zero as its emission factor; however, energy used for such purposes is not eligible as an SSC CDM Type I project component. | zero is used as its emission factor, and this part of biogas is not eligible as an SSC CDM Type I project component, which means no emission reductions will be claimed for this part of biogas utilization. |
| 6 | New facilities (Greenfield projects) and project activities involving capacity additions compared to the baseline scenario are only eligible if they comply with the related and relevant requirements in the "General guidelines for SSC CDM methodologies". | Applicable.<br><br>The project is a Greenfield project. The emission reduction sourced from methane recovery is 44,184 $tCO_2e/yr$, which is lower than the threshold of 60,000 $tCO_2e/yr$; the total installed capacity of the project is 3.83MW, which is lower than the threshold of 15MW.<br><br>Therefore, the Project is in line with" General Guidelines to SSC CDM methodologies" |
| 7 | The requirements concerning demonstration of the remaining lifetime of the replaced equipment shall be met as described in the "General guidelines for SSC CDM methodologies". | The project is a Greenfield project, does not involve the replaced equipment; therefore, this is irrelevant. |
| 8 | Measures are limited to those that result in aggregate emission reductions of less than or equal to 60 kt $CO_2$ equivalent annually from all Type III components of the project activity. | Applicable.<br><br>The emission reductions from the recovery and destruction of methane (viz. Type III components of the project) are 44,184 $tCO_2e/yr$, which is less than 60 kt $CO_2$ equivalent. |

Table 3-2 Applicability of AMS-I.D

| No. | Applicability conditions of the methodology | The project |
|---|---|---|
| 1 | This methodology comprises renewable energy generation units, such as photovoltaic, hydro, tidal/wave, wind, geothermal and renewable biomass:<br><br>(a) Supplying electricity to a national or a regional grid; or | Applicable.<br><br>The project will use biogas to generate electricity, and the electricity will be supplied to North China Power Grid. |

| | | |
|---|---|---|
| | (b) Supplying electricity to an identified consumer facility via national/regional grid through a contractual arrangement such as wheeling. | |
| 2 | Illustration of respective situations under which each of the methodology (i.e., "AMS-I.D.: Grid connected renewable electricity generation", "AMS-I.F.: Renewable electricity generation for captive use and mini-grid" and "AMS-I.A.: Electricity generation by the user) applies is included in the appendix. | The illustration of respective situations for AMS-I.D stated in the appendix is the same as applicability condition No. 1 above. No need for further justification. |
| 3 | This methodology is applicable to project activities that:<br><br>(a) Install a Greenfield plant;<br><br>(b) Involve a capacity addition in (an) existing plant(s);<br><br>(c) Involve a retrofit of (an) existing plant(s);<br><br>(d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or<br><br>(e) Involve a replacement of (an) existing plant(s). | Applicable.<br><br>The project is a greenfield plant. |
| 4 | Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology:<br><br>(a) The project activity is implemented in an existing reservoir with no change in the volume of reservoir;<br><br>(b) The project activity is implemented in an existing reservoir, where the volume of reservoir is increased and the power density of the project activity, as per definitions given in the project emissions section, is greater than 4 W/m2;<br><br>(c) The project activity results in new reservoirs and the power density of the power plant, as per definitions given in the project emissions section, is greater than 4 W/m2. | Irrelevant. The project is not a hydro power plant. |

| | | |
|---|---|---|
| 5 | If the new unit has both renewable and non-renewable components (e.g., a wind/diesel unit), the eligibility limit of 15 MW for a small-scale CDM project activity applies only to the renewable component. If the new unit co-fires fossil fuel, the capacity of the entire unit shall not exceed the limit of 15 MW. | Irrelevant. The project does not involve non-renewable components. |
| 6 | Combined heat and power (co-generation) systems are not eligible under this category. | According to the FSR, the only output for the project is electricity and the project does not involve co-generation system. |
| 7 | In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15 MW and should be physically distinct1 from the existing units. | Irrelevant. The project does not involve capacity addition. |
| 8 | In the case of retrofit, rehabilitation or replacement, to qualify as a small-scale project, the total output of the retrofitted, rehabilitated or replacement power plant/unit shall not exceed the limit of 15 MW. | Irrelevant. The project does not involve retrofit, rehabilitation or replacement. |
| 9 | In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to a grid, then the baseline for the electricity component shall be in accordance with procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration other applicable Type-I methodologies such as “AMS-I.C.: Thermal energy production with or without electricity” shall be explored. | Irrelevant. The project does not involve landfill gas, waste gas, wastewater treatment and agro-industries projects. |
| 10 | In case biomass is sourced from dedicated plantations, the applicability criteria in the tool | Irrelevant. The project does not involve biomass. |

| | | |
|---|---|---|
| | "Project emissions from cultivation of biomass" shall apply. | |

In addition, the project meets the applicability conditions of the applied tools applied in the PD as follows:

Table 3-3 Applicability of applied tools

| Tool | Applicability | The project |
|---|---|---|
| Tool to calculate the emission factor for an electricity system (version 07.0) | This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects) | Applicable.<br>The project utilizes biogas for power generation to supply electricity to NCPG that dominated by fossil-fuel plants. Therefore, the tool is applied to estimate the OM, BM and/or CM when calculating baseline emissions for the project. |
| | Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option II a and option II b. If option II a is chosen, the conditions specified in "Appendix 1: Procedures related to off-grid power generation" should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in | In the host country as off-grid power generation is not significant. Therefore, emission factor for the project electricity system is calculated only for the grid power plants. Thus, this applicability criterion is satisfied. |

| | | |
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| | the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity. | |
| | In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country. | Applicable.<br>The project electricity system is located in a non-Annex I country. |
| | Under this tool, the value applied to the $CO_2$ emission factor of biofuels is zero. | The project is not involved in biofuels. |
| Project and leakage emissions from anaerobic digesters (version 02.0) | The following sources of project emissions are accounted for in this tool:<br>(a) $CO_2$ emissions from consumption of electricity associated with the operation of the anaerobic digester;<br>(b) $CO_2$ emissions from consumption of fossil fuels associated with the operation of the anaerobic digester;<br>(c) $CH_4$ emissions from the digester (emissions during maintenance of the digester, physical leaks through the roof and side walls, and release through safety valves due to excess pressure in the digester); and<br>(d) $CH_4$ emissions from flaring of biogas. | Applicable.<br>All sources of project emissions have been accounted. |
| | The following sources of leakage emissions are accounted for in this tool: | Irrelevant. The project does not involve composting, anaerobic decay or anaerobic storage of digestate. |

| | | |
|---|---|---|
| | (a) $CH_4$ and $N_2O$ emission from composting of digestate;<br><br>(b) $CH_4$ emissions from the anaerobic decay of digestate disposed in a SWDS or subjected to anaerobic storage, such as in a stabilization pond. | |
| | Emission sources associated with $N_2O$ emissions from physical leakages from the digester, transportation of feed material and digestate or any other on-site transportation, piped distribution of the biogas, aerobic treatment of liquid digestate and land application of the digestate are neglected because these are minor emission sources or because they are accounted in the methodologies referring to this tool. | Applicable. As per the applied methodology, $N_2O$ emissions are neglected because these are minor emission sources. |
| Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation (version 03.0) | If emissions are calculated for electricity consumption, the tool is only applicable if one out of the following three scenarios applies to the sources of electricity consumption:<br><br>(a) Scenario A: Electricity consumption from the grid. The electricity is purchased from the grid only, and either no captive power plant(s) is/are installed at the site of electricity consumption or, if any captive power plant exists on site, it is either not operating or it is not physically able to provide electricity to the electricity consumer; | Applicable.<br><br>The source of electricity consumption of the project is Scenario A: Electricity consumption from the grid |

| | | |
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| | (b) Scenario B: Electricity consumption from (an) off-grid fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants are installed at the site of the electricity consumer and supply the consumer with electricity. The captive power plant(s) is/are not connected to the electricity grid; or<br><br>(c) Scenario C: Electricity consumption from the grid and (a) fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants operate at the site of the electricity consumer. The captive power plant(s) can provide electricity to the electricity consumer. The captive power plant(s) is/are also connected to the electricity grid. Hence, the electricity consumer can be provided with electricity from the captive power plant(s) and the grid. | |
| | This tool can be referred to in methodologies to provide procedures to monitor amount of electricity generated in the project scenario, only if one out of the following three project scenarios applies to the recipient of the electricity generated:<br><br>(a) Scenario I: Electricity is supplied to the grid;<br><br>(b) Scenario II: Electricity is supplied to consumers/electricity consuming facilities; or | Applicable.<br><br>The electricity generated by the project is supplied to the NCPG (Scenario I). |

| | | |
|---|---|---|
| | (c) Scenario III: Electricity is supplied to the grid and consumers/electricity consuming facilities. | |
| | This tool is not applicable in cases where captive renewable power generation technologies are installed to provide electricity in the project activity, in the baseline scenario or to sources of leakage. The tool only accounts for $CO_2$ emissions. | Applicable<br><br>No captive renewable power generation technologies are installed to provide electricity in the project activity, in the baseline scenario or to sources of leakage. |

## 3.3 Project Boundary

According to the methodology AMS-III.D, the project boundary includes the physical, geographical site(s) of (a) The livestock; (b) Animal manure management systems (including centralised manure treatment plant where applicable); (c) Facilities which recover and flare/combust or use methane.

According to the methodology AMS-I. D, the spatial extent of the project boundary includes the project power plant and all power plants connected physically to the electricity system that the CDM project power plant is connected to.

Hence, the project boundary of the Project includes the physical and geographical sites of the livestock, the centralized animal manure management system, the biogas power plant and all the power plants connected into the NCPG.

Figure 3-1 describes the project boundary of the Project Activity.

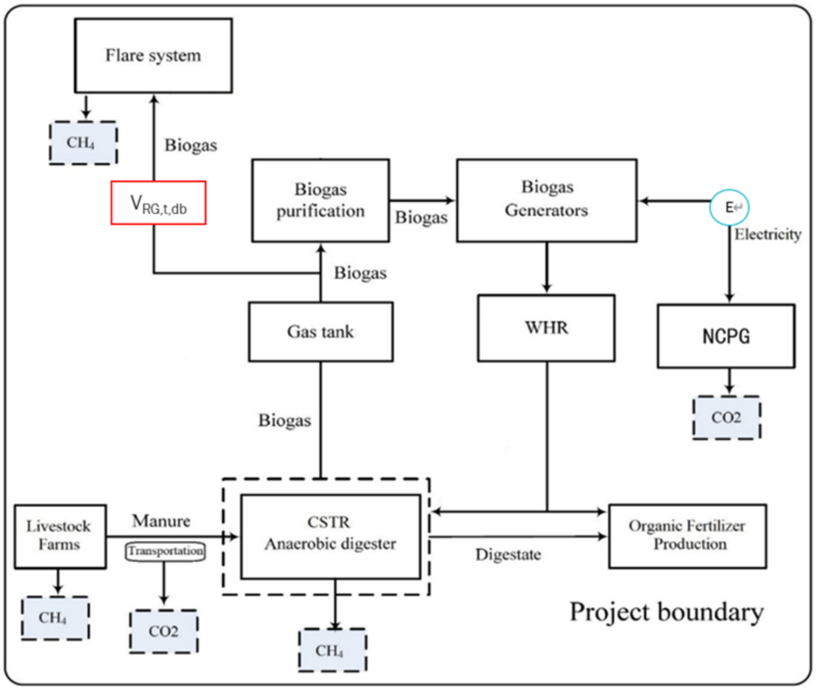


Figure 3-1 the project boundary of the Project Activity

Table 3-4 Emission sources included in or excluded from the project boundary

| Source | | Gas | Included? | Justification/Explanation |
|---|---|---|---|---|
| Baseline | Direct emissions from the manure treatment processes | $CO_2$ | No | Excluded for simplification. |
| | | $CH_4$ | Yes | The major source of emissions in the baseline |
| | | $N_2O$ | No | Excluded for simplification. |
| | Emissions from electricity consumption /Generation | $CO_2$ | Yes | The major source of emissions in the baseline |
| | | $CH_4$ | No | Excluded for simplification. |
| | | $N_2O$ | No | Excluded for simplification. |
| Project | Physical leakage of biogas in the manure management systems | $CO_2$ | No | Excluded for simplification. |
| | | $CH_4$ | Yes | Main emission source. |
| | | $N_2O$ | No | Excluded for simplification. |
| | | $CO_2$ | No | Excluded for simplification. |

| Source | | Gas | Included? | Justification/Explanation |
|---|---|---|---|---|
| | Emissions from flaring or combustion of the gas stream | $CH_4$ | No | Maybe a main emission resource[15]. |
| | | $N_2O$ | No | Excluded for simplification. |
| | Emissions from use of fossil fuels or electricity | $CO_2$ | Yes | The project consumes electricity during operation, so emission from use of electricity is the main emission source.<br>The project does not involve fossil fuel consumption, so the emission from use of fossil fuels is not included. |
| | | $CH_4$ | No | Excluded for simplification. |
| | | $N_2O$ | No | Excluded for simplification. |
| | Emissions from incremental transportation distances | $CO_2$ | No | Excluded for simplification. |
| | | $CH_4$ | No | Excluded for simplification. |
| | | $N_2O$ | No | Excluded for simplification. |
| | Emissions from the storage of manure | $CO_2$ | No | Excluded for simplification. |
| | | $CH_4$ | No | The storage time of the manure after removal from the animal barns, including transportation, is within 24 hours before being fed into the anaerobic digester, hence Emissions from the storage of manure is not accounted for. |
| | | $N_2O$ | No | Excluded for simplification. |

## 3.4 Baseline Scenario

As per para. 17 of AMS-III.D, the baseline scenario is the situation where, in the absence of the project activity, animal manure is left to decay anaerobically within the project boundary and methane is emitted to the atmosphere.

As per para. 19 of AMS-I. D, the baseline scenario is that the electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid.

[15] Only when emergency occurs, the emergency flare system will be used and the biogas will be flared. In this situation, there will be project emission from flaring.

Hence, the baseline scenario of the project is the animal manure waste was left to decay in anaerobic manure management system (lagoon) at the livestock farms and methane is emitted to the atmosphere directly without any methane recovery and destruction facility; equivalent amount of electricity was supplied from the NCPG connected fossil fuel fired power plants.

## 3.5 Additionality

As per para. 15-16 of applied methodology AMS-III.D, project activities may demonstrate the additionality by showing that there is no regulation in the host country, applicable to the project site, that requires the collection and destruction of methane from livestock manure. If so, it is not required to apply the “Guidelines on the demonstration of additionality of small-scale project activities”. This additionality condition also applies to Greenfield project activities. Furthermore, for project activities applying this methodology in combination with a Type I methodology, that has an energy component whose installed capacity is less than 5 MW, this procedure for additionality demonstration also applies to that component.

In line with AMS-III.D, the project recovers biogas to generate electricity with a total installed capacity of 3.83MW, which is lower than 5MW. The regulations relative to the project in China are identified as below.

a) Law of the People's Republic of China on Environment Protection;

b) Law of the People's Republic of China on the Prevention and Control of Solid Waste Pollution;

c) Law of the People's Republic of China on the Prevention and Control of Atmospheric Pollution;

d) Regulations on prevention and control of pollution from large scale livestock and poultry breeding;

e) Discharge standard of pollutants for livestock and poultry breeding (GB 18596-2001);

f) Technical standard of pollution prevention for livestock and poultry breeding (HJ/T 81-2001).

It has been identified that all the above laws and regulations in China **do not** require the collection and destruction of methane from livestock manure. In line with AMS-III.D, the project is deemed automatically additional.

## 3.6 Methodology Deviations

As plant specific fuel consumption and electricity generation data are not publicly available in China, the Guidance caused by DNV’s request for deviation of Chinese project activities for the CDM baseline methodology AM0005 has been applied for calculating the build margin (BM) emission factor of this project activity. The deviation has no impact on conservativeness of the quantification of GHG emission reductions or removals.

# 4 IMPLEMENTATION STATUS

## 4.1 Implementation Status of the Project Activity

The project is implemented in accordance with the description of this PD&MR in the former sections. The electricity generated using biogas, except a small portion for on-site usage, is supplied to NCPG through the nearby distribution network.

According to the feasibility study report, the intended installed capacity of the electricity generator is 3.83MW (1.165MW * 2 + 1.5MW *1). During this monitoring period, a set of internal combustion engine generators with an installed capacity of 2.33MW (1.165 MW *2) have been installed by the project owner. The installation of the other 1.5MW would depend on the circumstances in the near future. According to the present installed capacity of CSTR system and other supporting facilities, total design biogas output is 8,760,000 $m^3$ biogas per year, the same as the description of this PD&MR.

This is a well-known and high reliable technology for biogas utilization. High performance and reliability are guaranteed for the equipment. The annual average running time is expected to be 8,000 hours as determined by a certified design institute. In this monitoring period, electricity of 17,296.840 MWh generated by the project has been supplied to NCPG.

The timeline of the project is as below:

| | |
|---|---|
| Start date of construction | 07-Dec-2019 |
| Start date of operation | 21-Mar-2021 |
| Start date of the crediting period | |

Electricity meter was installed to monitor the electricity exported to NCPG. During this monitoring period, the power plant run continuously. No events or situations, which may impact the applicability of the applied methodology, occurred during this monitoring period.

# 5 ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

## 5.1 Baseline Emissions

According to the methodology AMS-III.D and AMS-I. D, baseline emissions are determined according to equation (1) and comprise the following sources:

$$BE_y = BE_{CH4,y} + BE_{Elec,y} \quad (1)$$

Where:

| | | |
|---|---|---|
| $BE_y$ | = | Baseline emissions in year *y* (t $CO_2e$/yr) |
| $BE_{CH4,y}$ | = | Baseline emissions from the manure treatment processes in year *y* (t $CO_2e$/yr) |
| $BE_{Elec,y}$ | = | Baseline emissions associated with electricity generation in year *y* (t $CO_2$/yr) |

**1. Baseline emissions from the manure treatment processes in year y ($BE_{CH4,y}$)**

As per AMS-III.D, Baseline emissions ($BE_{CH4,y}$) are calculated by using one of the following two options:

(a) Using the amount of the waste or raw material that would decay anaerobically in the absence of the project activity, with the most recent IPCC Tier 2 approach (please refer to the chapter 'Emissions from Livestock and Manure Management' under the volume 'Agriculture, Forestry and other Land use' of the 2006 IPCC Guidelines for National Greenhouse Gas Inventories). For this calculation, information about the characteristics of the manure and of the management systems in the baseline is required. Manure characteristics include the amount of volatile solids (VS) produced by the livestock and the maximum amount of methane that can be potentially produced from that manure (Bo);

(b) Using the amount of manure that would decay anaerobically in the absence of the project activity based on direct measurement of the quantity of manure treated together with its specific volatile solids (SVS) content.

The project applies option (a) to calculate baseline emissions from the manure management ($BE_y$), and baseline emissions are determined as follows:

$$BE_{CH4,y} = GWP_{CH4} \times D_{CH4} \times UF_b \times \sum_{j,LT} MCF_j \times B_{0,LT} \times N_{LT,y} \times VS_{LT,y} \times MS\%_{Bl,j} \quad (2)$$

Where:

| | | |
|---|---|---|
| $GWP_{CH4}$ | = | Global Warming Potential (GWP) of $CH_4$ applicable to the crediting period ($tCO_2e/tCH_4$) |
| $D_{CH4}$ | = | $CH_4$ density (0. 67kg/m$^3$ at room temperature (20ºC) and 1 atm pressure) |
| $UF_b$ | = | Model correction factor to account for model uncertainties (0.94) |
| LT | = | Index for all types of livestock |
| j | = | Index for animal manure management system |
| $MCF_j$ | = | Annual methane conversion factor (MCF) for the baseline animal manure management system j |
| $B_{0,LT}$ | = | Maximum methane producing potential of the volatile solid generated for animal type LT ($m^3CH_4$/kg-dm) |
| $N_{LT,y}$ | = | Annual average number of animals of type LT in year y (numbers) |
| $VS_{LT,y}$ | = | Volatile solids production/excretion per animal of livestock LT in year y (on a dry matter weight basis, kg-dm/animal/year) |
| $MS\%_{Bl,j}$ | = | Fraction of manure handled in baseline animal manure management system j |

a) The maximum methane-producing capacity of the manure ($B_{o,LT}$) varies by species and diet. The preferred method to obtain $B_{o,LT}$ measurement values is to use data from country-specific published sources, measured with a standardised method ($B_{o,LT}$ shall be based on total as-excreted VS). These values shall be compared to IPCC default values and any significant differences shall be explained. If country specific $B_o$ values are not available, default values from tables 10 A-4 to 10 A-9 of 2006 IPCC Guidelines for National Greenhouse Gas Inventories volume 4 Chapter 10 can be used, provided that the project participants assess the suitability of those data to the specific situation of the treatment site.

b) VS are the organic material in livestock manure and consist of both biodegradable and non-biodegradable fractions. For the calculations the total VS excreted by each animal species is required.

   If country specific VS values are not available IPCC default values from 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 Chapter 10 table 10 A-4 to 10 A-9 can be used provided that the project participants assess the suitability of those data to the specific situation of the treatment site particularly with reference to feed intake levels;

c) $B_{o,LT}$ values or VS values applicable to developed countries can be used provided the following four conditions are satisfied:

   i. The genetic source of the livestock originates from an Annex I Party;

ii. The farm uses formulated feed rations (FFR) which are optimized for the various animal(s), stage of growth, category, weight gain/productivity and/or genetics;

iii. The use of FFR can be validated (through on-farm record keeping, feed supplier, etc.);

iv. The project specific animal weights are more similar to developed country IPCC default values.

d) Methane Conversion Factors (MCF) values are determined for a specific manure management system and represent the degree to which $B_o$ is achieved. Where available country-specific MCF values that reflect the specific management systems used in particular countries or regions shall be used. Alternatively, the IPCC default values provided in table 10.17 of 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 Chapter 10 can be used. The site annual average temperature is taken from official data at the nearest meteorological station, or from data available from historical on-site observations.

e) The annual average number of animals ($N_{LT,y}$) is determined as follows:

$$N_{LT,y} = N_{da,y} \times (\frac{N_{p,y}}{365}) \quad (3)$$

Where:

| | | |
|---|---|---|
| $N_{da,y}$ | = | Number of days animal is alive in the farm in the year y (numbers) |
| $N_{p,y}$ | = | Number of animals produced annually of type LT for the year y (numbers) |

**2. Baseline emissions associated with electricity generation in year y (*$BE_{elec,y}$*)**

As per AMS-I. D, baseline emissions associated with electricity generation in year y is calculated according to equation (4) as below:

$$BE_{elec,y} = EG_{PJ,y} \times EF_{grid,y} \quad (4)$$

Where:

| | | |
|---|---|---|
| $EG_{PJ,y}$ | = | Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the project activity in year y (MWh) |
| $EF_{grid,y}$ | = | Combined margin $CO_2$ emission factor for grid connected power generation in year y calculated using the latest version of the "Tool to calculate the emission factor for an electricity system" (t $CO_2$/MWh) |

**Calculation of combined margin $CO_2$ emission factor for grid (*$EF_{grid,y}$*)**

According to "Tool to calculate the emission factor for an electricity system" (version 07.0). The following six steps are applied to determine OM, BM, and CM used for calculating the project baseline emissions:

- **Step 1:** Identify the relevant electricity systems;
- **Step 2:** Choose whether to include off-grid power plants in the project electricity system (optional);
- **Step 3:** Select a method to determine the operating margin (OM);
- **Step 4:** Calculate the operating margin emission factor according to the selected method;
- **Step 5:** Calculate the build margin (BM) emission factor;
- **Step 6:** Calculate the combined margin (CM) emission factor.

As China DNA has published the calculation method for emission factor of grid, the published data and method have been applied for this project to calculate operating margin (OM) and build margin, as following steps:

**Step 1: Identify the relevant electricity systems;**

This project site is in Hebei province of China, which belongs to North China Power Grid (NCPG) according to the public delineation of DNA[16], so NCPG is identified as the relevant electric system.

**Step 2: Choose whether to include off-grid power plants in the project electricity system (optional);**

For this project, Option I (only grid power plants are included in the calculation) is chosen.

**Step 3: Select a method to determine the operating margin (OM);**

Calculation of Operating Margin should be based on one of the four following methods according to the tool:

(a) Simple OM, or

(b) Simple adjusted OM, or

(c) Dispatch Data Analysis OM, or

(d) Average OM.

According to 'Methodological Tool: TOOL07 (Version 07.0) para40(a)(i), the Simple OM method is applicable to the project if the low-cost resources constitute less than 50% of total grid generation on average in the five most recent years. As the low-cost / must-run resources, i.e. $Share_{LCMR}$, is 7.71%[17] which constituted less than 50% of total electricity generation of NCPG in recent five years (10.10%, 8.94%, 7,21%,6,39% and 5,92% in 2017, 2016, 2015, 2014 and 2013,

[16]https://www.mee.gov.cn/ywgz/ydqhbh/wsqtkz/202012/W020201229610353340851.pdf

[17] 7.71% = (10.10%+8.94%+7.21%+6.39%+5.92%)/5

respectively)[18], the Simple OM (a) method is selected and the following data vintage is chosen to calculate the emission factor:

Ex ante option is chosen: As per tool: If the ex-ante option is chosen, the emission factor is determined once at the validation stage, thus no monitoring and recalculation of the emissions factor during the crediting period is required. For grid power plants, use a 3-year generation-weighted average, based on **the most recent data available** at the time of submission of the VCS-PDD to the VVB for validation, without requirement to monitor and recalculate the emissions factor during the crediting period. Thus, the emission factor is determined once at the validation stage and no monitoring and recalculation of the emissions factor during the crediting period is required.

**Step 4: Calculate the operating margin emission factor according to the selected method;**

The Simple OM emission factor ($EF_{OM,y}$) is calculated as the generation-weighted average emissions per electricity unit ($tCO_2e/MWh$) of all generating sources serving in the system, excluding low-operating cost and must-run power plants. It may be calculated:

Option A: Based on the net electricity generation and a $CO_2$ emission factor of each power plant / unit, or

Option B: Based on the total net electricity generation of all power plants serving the system and the fuel types and total fuel consumption of the project electricity system.

Option B can only be used if:

(a) The necessary data for Option A is not available; and

(b) Only nuclear and renewable power generation are considered as low-cost/must-run power sources and the quantity of electricity supplied to the grid by these sources is known; and

(c) Off-grid power plants are not included in the calculation.

In this project, all of the above conditions can be met, so Option B was chosen.

Under this option, the simple OM emission factor is calculated based on the net electricity supplied to the grid by all power plants serving the system, not including low-cost/must-run power plants/units, and based on the fuel type(s) and total fuel consumption of the project electricity system, as follows:

$$EF_{grid,OMsimple,y} = \frac{\sum_i (FC_{i,y} \cdot NCV_{i,y} \cdot EF_{CO_2,i,y})}{EG_y} \quad (5)$$

Where:

18 China Electric Power Yearbook (2014, 2015, 2016, 2017, 2018)

| | | |
|---|---|---|
| $EF_{grid,OMsimple,y}$ | = | Simple operating margin $CO_2$ emission factor in year y (t $CO_2$/MWh) |
| $FC_{i,y}$ | = | Amount of fuel type i consumed in the project electricity system in year y (mass or volume unit) |
| $NCV_{i,y}$ | = | Net calorific value (energy content) of fuel type i in year y (GJ/mass or volume unit) |
| $EF_{CO2,i,y}$ | = | $CO_2$ emission factor of fuel type i in year y (t $CO_2$/GJ) |
| $EG_y$ | = | Net electricity generated and delivered to the grid by all power sources serving the system, not including low-cost/must-run power plants/units, in year y (MWh) |
| i | = | All fuel types combusted in power sources in the project electricity system in year y |
| y | = | The relevant year as per the data vintage chosen in **Step 3**. |

Based on the most recent three years (2015-2017) where the data are the latest and available at the time of this PD submission, the calculation result of $EF_{grid,OM,,y}$ is 0.9419 $tCO_2e$/MWh. The data is published by Ministry of Ecology and Environment of the People's Republic of China[19].

**Step 5: Calculate the build margin (BM) emission factor;**

As per Section 6.5 of TOOL07 (version 07.0), in terms of vintage of data, project participants can choose between one of the following two options:

(a) Option 1 - for the first crediting period, calculate the build margin emission factor ex ante based on the most recent information available on units already built for sample group m at the time of CDM-PDD submission to the DOE for validation. For the second crediting period, the build margin emission factor should be updated based on the most recent information available on units already built at the time of submission of the request for renewal of the crediting period to the DOE. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used. This option does not require monitoring the emission factor during the crediting period;

[19] https://www.mee.gov.cn/ywgz/ydqhbh/wsqtkz/202012/t20201229_815386.shtml

(b) Option 2 - For the first crediting period, the build margin emission factor shall be updated annually, ex post, including those units built up to the year of registration of the project activity or, if information up to the year of registration is not yet available, including those units built up to the latest year for which information is available. For the second crediting period, the build margin emissions factor shall be calculated ex ante, as described in Option 1 above. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used.

In line with China's Baseline emission factors of regional grids 2019 ($BEF_{2019}$) published by the Development and Reform Commission of China, Option 1 is chosen for the project; the BM emission factor is calculated ex ante based on the most recent information available on units already built for sample group m at the time of this project description submission.

The sample group of power units m used to calculate the build margin should be determined as per the following procedure, consistent with the data vintage selected above:

(a) Identify the set of five power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently ($SET_{5\text{-units}}$) and determine their annual electricity generation ($AEG_{SET\text{-}5\text{-units}}$, in MWh);

(b) Determine the annual electricity generation of the proposed project electricity system, excluding power units registered as CDM project activities ($AEG_{total}$, in MWh). Identify the set of power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently and that comprise 20% of $AEG_{total}$ (if 20% falls on part of the generation of a unit, the generation of that unit is fully included in the calculation) ($SET_{\geq 20\%}$) and determine their annual electricity generation ($AEG_{SET\geq 20\%}$, in MWh);

(c) From $SET_{5\text{-units}}$ and $SET_{\geq 20\%}$ select the set of power units that comprises the larger annual electricity generation ($SET_{sample}$);

Identify the date when the power units in $SET_{sample}$ started to supply electricity to the grid. If none of the power units in $SET_{sample}$ started to supply electricity to the grid more than 10 years ago, then use $SET_{sample}$ to calculate the build margin. In this case ignore Steps (d), (e) and (f).

Otherwise:

(d) Exclude from $SET_{sample}$ the power units which started to supply electricity to the grid more than 10 years ago. Include in that set the power units registered as CDM project activities, starting with power units that started to supply electricity to the grid most recently, until the electricity generation of the new set comprises 20% of the annual electricity generation of the proposed project electricity system (if 20% falls on part of the generation of a unit, the generation of that unit is fully included in the calculation) to the extent is possible. Determine for the resulting set ($SET_{sample\text{-}CDM}$) the annual electricity generation ($AEG_{SET\text{-}sample\text{-}CDM}$, in MWh); If the annual electricity generation of that set comprises at least 20% of the annual electricity generation of the proposed project electricity system (i.e. $AEG_{SET\text{-}sample\text{-}CDM} \geq 0.2 \times AEG_{total}$), then use the sample group $SET_{sample\text{-}CDM}$ to calculate the build margin. Ignore steps (e) and (f).

Otherwise:

(e) Include in the sample group $SET_{sample\text{-}CDM}$ the power units that started to supply electricity to the grid more than 10 years ago until the electricity generation of the new set comprises 20% of the annual electricity generation of the proposed project electricity system (if 20% falls on part of the generation of a unit, the generation of that unit is fully included in the calculation);

(f) The sample group of power units m used to calculate the build margin is the resulting set ($SET_{sample\text{-}CDM\text{->}10yrs}$).

The BM emissions factor is the generation-weighted average emission factor ($tCO_2/MWh$) of all power units m during the most recent year y for which electricity generation data is available, calculated as follows:

$$EF_{grid,BM,y} = \frac{\sum_m (EG_{m,y} \times EF_{EL,m,y})}{\sum_m EG_{m,y}} \quad (6)$$

where:

| | | |
|---|---|---|
| $EF_{grid,BM,y}$ | = | Build margin emission factor of NCPG ($tCO_2/MWh$) |
| $EG_{m,y}$ | = | Net quantity of electricity generated and delivered to the grid by power unit m in year y (MWh) |
| $EF_{EL,m,y}$ | = | $CO_2$ emission factor of power unit m in year y ($t\ CO_2/MWh$) |
| $m$ | = | Power units included in the build margin |
| $y$ | = | The most recent year for which the generation data is available |

As it is difficult to obtain the detailed data on the power generation, fuel consumption and thermal efficiency of each newly built power unit from public documents, a deviation of TOOL07 (07.0) is adopted following the clarifications[20] given by the CDM EB concerning the BM emission factor calculation:

(1) The CDM EB suggested using the efficiency level of the best technology commercially available in the provincial/regional or national grid of China, as a conservative proxy, for each fuel type in estimating the fuel consumption to estimate the build margin.

[20] "Request for clarification on use of approved methodology AM0005 for several projects in China", the EB's guidance on DNV deviation request. http://cdm.unfccc.int/UserManagement/FileStorage/AM_CLAR_QEJWJEF3CFBP1OZAK6V5YXPQKK7WYJ

(2) The EB agreed the use of capacity additions during last 1 ~ 3 years for estimating the build margin emission factor for grid electricity.

(3) The EB also agreed to use of weights estimated using installed capacity in place of annual electricity generation.

The newly built power plants in the past few years are bundled into "grouped new power plant" according to their construction year, their province and their fuel type. The annual net electricity generation in the year y of each "grouped new power plant" $EG_{m,y}$ is estimated according to their total capacity and the average utilization hours, as the following equation:

$$EG_{m,y} = CAP_m \times H_{m,y} \quad (7)$$

where:

| | | |
|---|---|---|
| $EG_{m,y}$ | = | Annual net electricity generation the unit m in year y (MWh) |
| $CAP_m$ | = | Installed capacity of the unit m (MW) |
| $H_{m,y}$ | = | Utilization hour of the unit m in the year y (h), determined according to the average utilization hour of the same type of unit in the same province |
| $y$ | = | The most recent year for which the generation data is available. For the calculation of BM in 2019, y = 2017 |
| $m$ | = | grouped new power plant |

Since the newly built power plants in the same province (A), in the same year (t) and using the same fuel type (k) are grouped into "a grouped new power plant", $CAP_m$ represents the total installed capacity of fuel type k power plants located in the provinces A and in the year t:

$$CAP_m = CAP_{A,t,k} \quad (8)$$

where:

| | | |
|---|---|---|
| $CAP_m$ | = | Installed capacity of the unit m (MW), with m representing the specified combination of A, t, and k |
| $CAP_{A,t,k}$ | = | Total installed capacity of fuel type k power plants located in the province A and in the year t |
| $A$ | = | Provinces covered by the NCPG, namely, Beijing City, Tianjin City, Hebei Province, Shanxi Province, Shandong Province and Inner Mongolia Autonomous Region |

| | | |
|---|---|---|
| $t$ | = | Years related to the grouped new power plants, for the 2019 calculation, t represents 2017, 2016, 2015.... Until the aggregated electricity generation of the grouped new power plants reaches 20% of the total electricity generation of the NCPG |
| $k$ | = | Fuel type of the grouped new power plants, including hydro, thermal (coal, gas, oil, waste incineration, other thermal), nuclear, wind, solar and others. |

Figure 4 shows the procedure to determine the sample group of power units m.

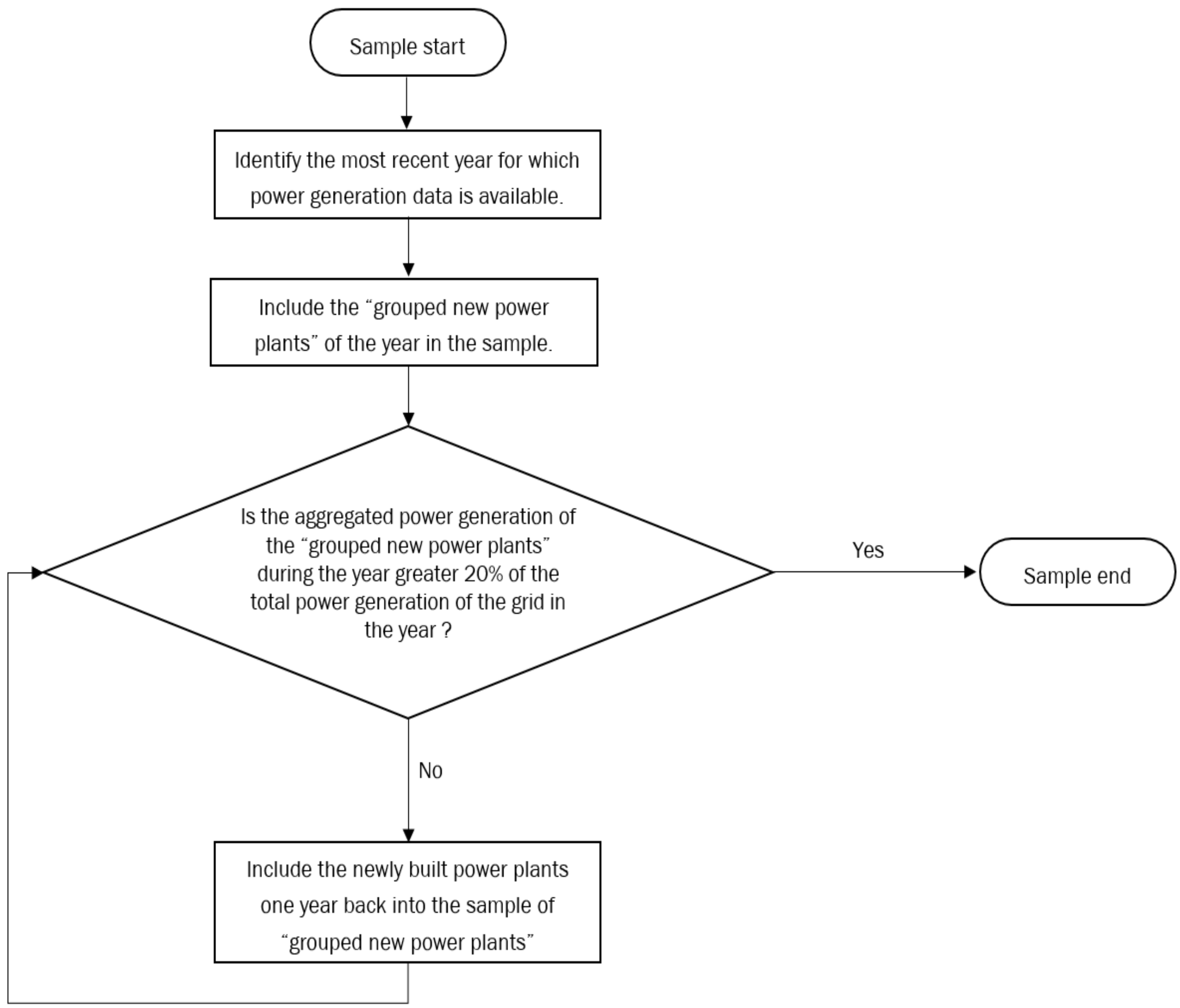


Figure 4 Procedure to determine the sample group of power units m used for the BM emission factor calculation

The emission factors of each fuel type $EF_{EL,m,y}$ are determined according to the Option A2 in the TOOL07, as the following equation:

$$EF_{EL,m,y} = \frac{EF_{CO2,m,i,y} \times 3.6}{\eta_{m,y}} \tag{9}$$

where:

| | | |
|---|---|---|
| $EF_{EL,m,y}$ | = | $CO_2$ emission factor of power unit m in year y (t $CO_2$/MWh) |
| $EF_{CO2,m,i,y}$ | = | Average $CO_2$ emission factor of fuel type i used in power unit m in year y (t $CO_2$/GJ) |
| $\eta_{m,y}$ | = | Average net energy conversion efficiency of power unit m in year y (ratio) |
| $m$ | = | All power units serving the grid in year y except low-cost / must-run power units |
| 3.6 | = | Conversion factor (GJ/MWh) |

Among the fuel types, the emission factors of hydro, nuclear, wind, solar, other thermal and others are 0. Concerning the emission factors of coal, gas, oil and waste incineration, Equation takes the following form due to conservativeness:

$$EF_{best,m,y} = \frac{EF_{CO2,m,i,y} \times 3.6}{\eta_{best,y}} \tag{10}$$

where:

| | | |
|---|---|---|
| $EF_{best,m,y}$ | = | Emission factor of power unit m with the best technology commercially available in year y (t $CO_2$/MWh) |
| $\eta_{best,y}$ | = | Power generation efficiency of the best technology commercially available in year y |
| $m$ | = | Power units serving the grid with coal, gas, oil or waste incineration in year y |

According to the latest and available data at the time of this PD submission, $EF_{grid,BM,y}$ is calculated to be 0.4819 $tCO_2e/MWh$. The data is published by Ministry of Ecology and Environment of the People's Republic of China[21].

**Step 6: Calculate the combined margin (CM) emission factor.**

The calculation of the combined margin emission factor ($EF_{grid,CM,y}$) is based on one of the following methods:

(a) Weighted average CM; or

(b) Simplified CM.

The weighted average CM method (option A) should be used as the preferred option.

The simplified CM method (option b) can only be used if:

a) The project activity is located in: (i) a Least Developed Country (LDC); or in (ii) a country with less than 10 registered CDM projects at the starting date of validation; or (iii) a Small Island Developing States (SIDS); and

b) The data requirements for the application of step 5 above cannot be met.

This PD choose option A.

The combined margin emissions factor is calculated as follows:

$$EF_{grid,CM,y} = EF_{grid,OM,y} \times w_{OM} + EF_{grid,BM,y} \times w_{BM} \quad (11)$$

Where:

| | | |
|---|---|---|
| $EF_{grid,OM,y}$ | = | operating margin emission factor of NCPG ($tCO_2e/MWh$) |
| $EF_{grid,BM,y}$ | = | build margin $CO_2$ emission factor of NCPG ($tCO_2e/MWh$) |
| $w_{OM}$ | = | the weighting of operating margin emission factor (%) |
| $w_{BM}$ | = | the weighting of build margin emission factor (%) |

According to the tool, for all other projects (except wind and solar power generation project activities): $w_{OM}$ = 0.5 and $w_{BM}$ = 0.5 for the first crediting period, and $w_{OM}$ = 0.25 and $w_{BM}$ = 0.75 for the second and third crediting period, unless otherwise specified in the approved methodology which refers to this tool.

[21] https://www.mee.gov.cn/ywgz/ydqhbh/wsqtkz/202012/t20201229_815386.shtml

As a wastes recycling project, $w_{OM}$ = 0.5 and $w_{BM}$ = 0.5 for the first crediting period.

$EF_{grid,CM,,y}$ = 0.9419 * 0.5 + 0.4819 * 0.5 = 0.7119 $tCO_2e/MWh$.

## 5.2 Project Emissions

According to the methodology AMS-III.D and AMS-I. D, project emissions consist of:

(a) Physical leakage of biogas in the manure management systems which includes production, collection and transport of biogas to the point of flaring/combustion or gainful use ($PE_{PL,y}$);

(b) Emissions from flaring or combustion of the gas stream ($PE_{flare,y}$);

(c) $CO_2$ emissions from use of fossil fuels or electricity for the operation of all the installed facilities ($PE_{power,y}$);

(d) $CO_2$ emissions from incremental transportation distances;

(e) Emissions from the storage of manure before being fed into the anaerobic digester ($PE_{storage,y}$).

$$PE_y = PE_{PL,y} + PE_{flare,y} + PE_{power,y} + PE_{transp,y} + PE_{storage,y} \tag{12}$$

Where:

| | | |
|---|---|---|
| $PE_y$ | = | Project emissions in year *y* (t $CO_2e$) |
| $PE_{PL,y}$ | = | Emissions due to physical leakage of biogas in year *y* (t $CO_2e$) |
| $PE_{flare,y}$ | = | Emissions from flaring or combustion of the biogas stream in the year *y* (t $CO_2e$) |
| $PE_{power,y}$ | = | Emissions from the use of fossil fuel or electricity for the operation of the installed facilities in the year *y* (t $CO_2e$) |
| $PE_{transp,y}$ | = | Emissions from incremental transportation in the year *y* (t $CO_2e$), as per relevant paragraph in AMS-III.AO |
| $PE_{storage,y}$ | = | Emissions from the storage of manure (t $CO_2e$) |

**1. Emissions due to physical leakage of biogas in year y**

Project emissions due to physical leakage of biogas from the animal manure management systems used to produce, collect and transport the biogas to the point of flaring or gainful use are estimated as:

$$PE_{PL,y} = 0.10 \times GWP_{CH4} \times D_{CH4} \times \sum_{i,LT} B_{0,LT} \times N_{LT,y} \times VS_{LT,y} \times MS\%_{i,y} \tag{13}$$

Where:

| | | |
|---|---|---|
| $GWP_{CH4}$ | = | Global Warming Potential (GWP) of $CH_4$ applicable to the crediting period (t $CO_2e$/t $CH_4$) |
| $D_{CH4}$ | = | $CH_4$ density (0.00067 t/m$^3$ at room temperature (20 ºC) and 1 atm pressure) |
| $LT$ | = | Index for all types of livestock |
| $i$ | = | Index for animal manure management system |
| $N_{LT,y}$ | = | Annual average number of animals of type LT in year y (numbers) |
| $VS_{LT,y}$ | = | Volatile solids production/excretion per animal of livestock LT in year y (on a dry matter weight basis, kg-dm/animal/year) |
| $MS\%_{i,y}$ | = | Fraction of manure handled in system i in year y. If the project activity involves sequential manure management systems, the procedure specified in paragraph 18(e) of AMS-III.D shall be used to estimate the project emissions due to physical leakage of biogas in each stage |
| $B_{0,LT}$ | = | Maximum methane producing potential of the volatile solid generated for animal type $LT$ (m$^3$ $CH_4$/kg-VS), as per paragraph18 of AMS-III.D |

**2. Emissions from flaring or combustion of the biogas stream in the year y**

According to methodology, in the case of flaring of the recovered biogas, project emissions are estimated using the procedures described in the methodological tool "Project emissions from flaring". If the recovered biogas is combusted for electrical/thermal energy production or for other gainful use, the methane destruction efficiency can be considered as 100%. However, this use of the recovered biogas shall be included in the project boundary and its output shall be monitored in order to ensure that the recovered biogas is actually destroyed, even if the emission reductions from this component are not claimed.

For the project, only when emergency occurs, the emergency flare system will be used and the biogas will be flared. The project emission is calculated using the procedures described in the methodological TOOL06 "Project emissions from flaring" (version 04.0) as below:

$$PE_{flare,\,y} = GWP_{CH4} \times \sum_{m=1}^{525600} F_{CH4,RG,m} \times (1 - \eta_{flare,\,m}) \times 10^{-3} \quad (14)$$

Where:

$PE_{flare,\,y}$ = Project emissions from flaring of the residual gas in year y ($tCO_2e$)

$GWP_{CH4}$ = Global warming potential of methane valid for the commitment period ($tCO_2e/tCH_4$)

$F_{CH4,\,RG,\,m}$ = Mass flow of methane in the residual gas in the minute m (kg)

$\eta_{flare,\,m}$ = Flare efficiency in the minute m

According to **Option A** in the "Tool to determine the mass flow of a greenhouse gas in a gaseous stream" (version 03.0), the calculation equation of mass flow of residual gas is listed directly as follows:

$$F_{CH4,RG,m} = V_{RG,t,db} \times \nu_{RG,t,db} \times \rho_{RG,n} \times \frac{h}{1000} \quad (15)$$

With:

$$\rho_{RG,n} = \frac{P_n \times MM_i}{R_u \times T_n} \quad (16)$$

Where:

| | | |
|---|---|---|
| $F_{CH4,RG,m}$ | = | Mass flow of greenhouse gas I in the in the residual gaseous stream in time interval t (kg gas/m) |
| $V_{RG,t,db}$ | = | Volumetric flow of the gaseous stream in time interval t on a dry basis (m³ dry gas/h) |
| $\nu_{RG,t,db}$ | = | Volumetric fraction of greenhouse gas i in the gaseous stream in a time interval t on a dry basis (m³ gas i/m³ dry gas) , i =Residual gas (RG), $\nu_{RG,t,db}$=$\nu_{i,t,db}$ |
| $\rho_{RG,n}$ | = | Density of greenhouse gas i in the gaseous stream in time interval t (kg gas i/m³ gas i), i =Residual gas (RG) |
| $P_n$ | = | Absolute pressure of the gaseous stream in time interval t (Pa) |
| $MM_i$ | = | Molecular mass of greenhouse gas i (kg/kmol), i =Residual gas (RG) |
| $R_u$ | = | Universal ideal gases constant (Pa.m³ /kmol.K) |
| $T_n$ | = | Temperature of the gaseous stream in time interval t (K) |
| h | = | The total operation hours of the flare (hours) |

Since this is a small scale project, a default value for the fraction of methane in the biogas ($f_{CH4,default}$) is applied to substitute the parameter $\nu_{RG,t,db}$. And $F_{CH4,\ RG,\ m}$ will be monitored ex post in the verification period.

Based on the FSR, all the methane generated by the project will be used as energy supply. In normal circumstances, the project proponent would like to consider a methane destruction efficiency of 100% of the generation system since all the biogas will be utilized and there will be no excess biogas flared. In order to ensure no biogas is released under exigencies, an enclosed emergency flare system is installed at the project site, this emergency flare system is not used under normal operation.

In the case of enclosed flares, project participants may choose between the following two options to determine the flare efficiency for minute m ($\eta_{flare,m}$) and shall document in the VCS-PD which option is selected: (a) Option A: Apply a default value for flare efficiency; (b) Option B: Measure the flare efficiency. The project applied Option A to determine the value of $\eta_{flare,m}$.

In case this emergency flare system operated, the $PE_{flare,y}$ will be monitored and calculated ex post. The estimated $PE_{flare,y}$ is 0 $tCO_2e$ ex ante.

**3. Emissions from the use of fossil fuel or electricity for the operation of the installed facilities in the year y**

Project emissions from electricity and fossil fuel consumption are determined by following the methodological tool "Project and leakage emissions from anaerobic digesters" (version 02.0), where $PE_{Power,y}$ is the sum of $PE_{EC,y}$ and $PE_{FC,y}$ in the tool.

The project does not use any fossil fuel, so $PE_{FC,y}$ is not included in the project. Thus, $PE_{FC,y}$ = 0 $tCO_2e$.

According to methodological tool "Project and leakage emissions from anaerobic digesters", $PE_{EC,y}$ shall be calculated using the "Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation" (version 03.0) as follows:

$$PE_{EC,y} = \sum_j EC_{PJ,j,y} \times EF_{EL,j,y} \times (1 + TDL_{j,y}) \quad (17)$$

Where:

| | | |
|---|---|---|
| $PE_{EC,y}$ | = | Project emissions from electricity consumption in year y (t $CO_2$ / yr) |
| $EC_{PJ,j,y}$ | = | Quantity of electricity consumed by the project electricity consumption source j in year y (MWh/yr) |
| $EF_{EL,j,y}$ | = | Emission factor for electricity generation for source j in year y (t $CO_2$/MWh) |
| $TDL_{j,y}$ | = | Average technical transmission and distribution losses for providing electricity to source j in year y |
| $j$ | = | Sources of electricity consumption in the project |

As stated above, according to the tool "Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation"(version 03.0), for the project is in the case of Scenario A: Electricity consumption from the grid, the project participants choose Option A1: Calculate the combined margin emission factor of the applicable electricity system, using the procedures in the latest approved version of the "Tool to calculate the emission factor for an electricity system" (version 07.0) ($EF_{EL,j,y} = EF_{grid,CM,y}$).

According to the tool "Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation" (version 03.0), the project belongs to the case of Scenario A, use as default values of 20% for project emission, i.e. $TDL_{j,y}$ =20%.

$EC_{PJ,j,y}$ is ex ante determined as 0 in the PD and will be monitored ex post in the verification period.

**4. Emissions from incremental transportation in the year y**

According to AMS-III.D, emission from transportation of animal manure between livestock farms and the project site is calculated using relevant paragraph in AMS-III.AO as below:

$PE_{transp,y} = (Q_y/CT_y)*DAF_w*EF_{CO2,transport} + (Q_{res\cdot waste,y}/CT_{res\cdot waste,y})*DAF_{res\cdot waste}*EF_{CO2,transport}$ (18)

Where:

| | | |
|---|---|---|
| $Q_y$ | = | Quantity of raw waste/manure treated and/or wastewater co-digested in the year y(tonnes) |
| $CT_y$ | = | Average truck capacity for transportation (tonnes/truck) |
| $DAF_w$ | = | Average incremental distance for raw solid waste/manure and/or wastewater transportation (km/truck) |
| $EF_{CO2,\ transport}$ | = | $CO_2$ emission factor from fuel use due to transportation ($kgCO_2$/km, IPCC default values or local values may be used) |
| $Q_{res\cdot waste,y}$ | = | Quantity of residual waste produced in year y (tonnes) |
| $CT_{res,waste,y}$ | = | Average truck capacity for residual waste transportation (tonnes/truck) |
| $DAF_{res\cdot waste}$ | = | Average distance for residual waste transportation (km/truck) |

The project site is at the east side of the pasture and waste from the pasture will all enter the receiving tank of the pre-treatment system through tight pipeline, thus there is no emission from transportation of animal manure between livestock farms and the project site. $PE_{transp,y}$ = 0.

**5. Emissions from the storage of manure**

Project emissions on account of storage of manure before being fed into the anaerobic digester shall be accounted for if both condition (a) and condition (b) below are satisfied:

(a) The storage time of the manure after removal from the animal barns, including transportation, exceeds 24 hours before being fed into the anaerobic digester;

(b) The dry matter content of the manure when removed from the animal barns is less than 20%.

The storage time of the manure after removal from the animal barns, including transportation, is 5~7 hours, which is within 24 hours before being fed into the anaerobic digester. Hence emissions from the storage of manure is not accounted for, $PE_{storage,y}$ = 0 $tCO_2e$.

## 5.3 Leakage

According to AMS-III.D Methane recovery in animal manure management systems (version 21.0): Leakage is determined by following the relevant procedure in the methodological tool “Project and leakage emissions from anaerobic digesters”.

As per “Project and leakage emissions from anaerobic digesters” (version 02.0): The following sources of leakage emissions are accounted for in this tool:

(a) $CH_4$ and $N_2O$ emission from composting of digestate;

(b) $CH_4$ emissions from the anaerobic decay of digestate disposed in a SWDS or subjected to anaerobic storage, such as in a stabilization pond.

The leakage emissions associated with the anaerobic digester depend on how the digestate is managed. The leakage emissions include emissions associated with storage of digestate and emissions associated with composting of the digestate.

The digestate produced after anaerobic fermentation is separated by solid-liquid separation, and the solid part is squeezed twice before entering the drying drum. The dried part is used as cow bedding material, and the liquid part is treated with aerobic aeration before being returned to farmland for fertilizer. The liquid part will be sent to aerobic treatment as soon as digestate is separated by solid-liquid separation. Digestate will not be stored under anaerobic conditions. In addition, the project does not involve anaerobic composting of digestate. Therefore, leakage emissions of the project associated with the anaerobic digester is not accounted for, leakage emission is 0

## 5.4 Estimated Net GHG Emission Reductions and Removals

The emission reductions achieved by the project activity will be determined ex post through direct measurement of the amount of methane fueled, flared or gainfully used. It is likely that the project activity involves manure treatment steps with higher methane conversion factors (MCF) than the MCF for the manure treatment systems used in the baseline situation, therefore the emission reductions achieved by the project activity are limited to the ex post calculated baseline emissions minus the project emissions using the actual monitored data for the project activity (i.e. $N_{LT}$,y, $MS\%_{i,y}$, as well as $VS_{LT,y}$ in cases where adjusted values for animal weight are used). The emission reductions achieved in any year are the lowest value of the following:

$$ER_{y,ex\ post} = \min\left[\left(BE_{y,ex\ post} - PE_{y,ex\ post}\right), \left(MD_y - PE_{power,y,ex\ post}\right)\right] \quad (19)$$

Where:

| | | |
|---|---|---|
| $ER_{y,ex\ post}$ | = | Emission reductions achieved by the project activity based on monitored values for year y ($tCO_2e$) |
| $BE_{y,ex\ post}$ | = | Baseline emissions calculated using equation 1. For projects using option in paragraph 17(a) using ex post monitored values of $N_{LT,y}$ and if applicable $VS_{LT,y}$. For projects using option in paragraph 17(b), the ex post monitored values for $Q_{manure,j,LT,y}$ and $SVS_{j,LT,y}$ are used |
| $PE_{y,ex\ post}$ | = | Project emissions calculated using equation 11 using ex post monitored values of $N_{LT,y}$, $MS\%_{i,y}$, and if applicable $VS_{LT,y}$ |
| $MD_y$ | = | Methane captured and destroyed or used gainfully by the project activity in year *y* ($tCO_2e$) |
| $PE_{power,y,ex\ post}$ | = | Emissions from the use of fossil fuel or electricity for the operation of the installed facilities based on monitored values in the year *y* ($tCO_2e$) |

As per para.30 of AMS-III.D, if project activities utilize the recovered methane for power generation, methane captured and destroyed or used gainfully by the project activity may be calculated as follows, based on the amount of monitored electricity generation, without monitoring methane flow and concentration:

$$MD_y = \frac{EG_y \times 3600}{NCV_{CH4} \times EE_y} \times D_{CH4} \times GWP_{CH4} \quad (20)$$

Where:

| | | |
|---|---|---|
| $EG_y$ | = | Total electricity generated from the recovered biogas in year *y* (MWh) |
| $3600$ | = | Conversion factor (1 MWh = 3600 MJ) |
| $NCV_{CH4}$ | = | NCV of methane (MJ/Nm$^3$) (use default value: 35.9 MJ/Nm$^3$) |
| $EE_y$ | = | Energy conversion efficiency of the project equipment, which is determined by adopting one of the following criteria:<br>Specification provided by the equipment manufacture. The equipment shall be designed to utilize biogas as fuel, and efficiency specification is for this fuel. If the specification provides a range of efficiency values, the highest value of the range shall be used for the calculation;<br>Default efficiency of 40 %. |

**1. Calculation of baseline emissions:**

Table 5-1: Ex-ante value of parameters to calculate $BE_y$

| Parameter | Value | Data sources |
|---|---|---|
| $GWP_{CH4}$ | 28 $tCO_{2e}/tCH_4$ | IPCC Fifth Assessment Report (AR5) |
| $D_{CH4}$ | 0.67 kg/m$^3$ | AMS-III.D |
| $UF_b$ | 0.94 | AMS-III.D |
| $MCF_j$ | 76% | Table 10.17 of 2019 refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 Chapter 10 |
| $B_{o,LT}$ | Dairy cattle:0.13 m$^3$$CH_4$/kg-VS | Table 10.16 of 2019 refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 Chapter 10 |
| $VS_{default}$ | Dairy cattle: 3.474 kg/hd/day | $VS_{default}$ =$VS_{rate(T)}$ * TAM / 1000 = 9.0 * 386 / 1000<br>TAM:Default Values for livestock weight for Animal Categories in Asia area is 386kg(mean) for dairy cattle |

| | | |
|---|---|---|
| | | $VS_{rate(T)}$ = Default Values for Volatile Solid Excretion Rate, 9.0 kg VS 1000 kg animal mass$^{-1}$ day$^{-1}$ for dairy cattle sourced from Volume 4, chapter 10, table 10.13A of 2019 Refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories. |
| $MS\%_{Bl,j}$ | 100% | FSR |
| $nd_y$ | 365 days | FSR |
| $N_{da,y}$ | 365 days | FSR |
| $N_{p,y}$ | 20,000 | FSR |

As per equations (1), (2), the result of the ex-ante calculated baseline emissions of methane from the manure treatment processes of the project is:

Table 5-2: Ex-ante Baseline Emissions $BE_y$

| Index for all types of livestock (LT) | The annual average number of animals ($N_{LT,y}$) | Volatile solids production of livestock LT in year y ($VS_{LT,y}$) (kg-dm/animal/year) | Baseline Emissions ($BE_y$) $tCO_2e$ |
|---|---|---|---|
| - | $N_{LT,y} = N_{da,y} \times (\frac{N_{p,y}}{365})$ | $VS_{LT,y} = VS_{default} \times nd_y$ | $BE_y = GWP_{CH4} \times D_{CH4} \times UF_b \times \sum_{j,LT} MCF_j \times B_{0,LT} \times N_{LT,y} \times VS_{LT,y} \times MS\%_{Bl,j}$ |
| Dairy cattle | 20,000 | 1,268.01 | 44,184 |
| Total | - | - | 44,184 |

As pre the FSR, the annual exported electricity to the NCPG of the project is expected to be 27,000 MWh, the combined margin $CO_2$ emission factor for the grid is 0.7119 $tCO_2$/MWh, hence the expected annual baseline emissions associated with electricity generation of the project is calculated as follows:

$BE_{elec,y} = EG_{PJ,y} \times EF_{grid,y}$ = 27,000 MWh× 0.7119 $tCO_2$/MWh = 19,221 $tCO_2$

$BE_y = BE_{CH4,y} + BE_{elec,y}$ = 44,184 $tCO_2e$ + 19,221 $tCO_2e$ = 63,405$tCO_2e$

**2. Calculation of project emissions**

Table 5-3: Ex-ante value of parameters to calculate project emissions

| Parameter | Value | Data sources |
|---|---|---|

| | | |
|---|---|---|
| $GWP_{CH4}$ | 28 $tCO_2e/tCH_4$ | IPCC Fifth Assessment Report (AR5) |
| $D_{CH4}$ ($\rho_{CH4}$) | 0.67 kg/m³ | AMS-III.D |
| $B_{o,LT}$ | Dairy cattle:0.13 $m^3CH_4/kg$-VS | Table 10.16 of 2019 refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 Chapter 10 |
| $VS_{default}$ | Dairy cattle: 3.474 kg/hd/day | $VS_{default}$ =$VS_{rate(T)}$ * TAM / 1000 = 9.0 * 386 / 1000<br>TAM:Default Values for livestock weight for Animal Categories in Asia area is 386kg(mean) for dairy cattle<br>$VS_{rate(T)}$ = Default Values for Volatile Solid Excretion Rate, 9.0 kg VS 1000 kg animal mass$^{-1}$ day$^{-1}$ for dairy cattle sourced from Volume 4, chapter 10, table 10.13A of 2019 Refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories. |
| $MS\%_{i,y}$ | 100% | FSR |
| $N_{da,y}$ | 365 days | FSR |
| $N_{p,y}$ | 20,000 | FSR |
| $f_{CH4,\ default}$ | 0.6 | "Tool to Project and leakage emissions from anaerobic digesters" |

### 2.1 Emissions due to physical leakage of biogas in year y

As described in 5.2, project emissions from physical leakage of biogas are calculated as per equations (12), (13).

Table 5-4: Ex-ante calculate Physical leakage of biogas ($PE_{PL,y}$)

| Index for all types of livestock (LT) | The annual average number of animals ($N_{LT,y}$) | Volatile solids production of livestock LT in year y ($VS_{LT,y}$) (kg-dm/animal/year) | Physical leakage of biogas ($PE_{PL,y}$) $tCO_2e$ |
|---|---|---|---|
| - | $N_{LT,y} = N_{da,y} \times (\frac{N_{p,y}}{365})$ | $VS_{LT,y} = VS_{default} \times nd_y$ | $PE_{PL,y} = 0.10 \times GWP_{CH4} \times D_{CH4} \times \sum_{j,LT} B_{0,LT} \times N_{LT,y} \times VS_{LT,y} \times MS\%_{i,y}$ |
| Dairy cattle | 20,000 | 1,268.01 | 6,185 |
| Total | - | - | 6,185 |

So, the ex-ante calculated result of the project's Physical leakage of biogas is 6,185 $tCO_2e$.

**2.2 Emissions from flaring or combustion of the biogas stream in the year y**

As described above, in case this emergency flare system operated, the $PE_{flare,y}$ will be monitored and calculated ex post. The estimated $PE_{flare,y}$ is 0 $tCO_2e$ ex ante.

**2.3 Emissions from the use of fossil fuel or electricity for the operation of the installed facilities in the year y**

As described in 5.2, the project does not use any fossil fuel, so $PE_{FC,y}$ is not included in the project emission.

As described in 5.2, the electricity consumption of the project site is ex ante determined as 0 in the PD and will be monitored ex post in the verification period. Thus, $PE_{EC,y}$ = 0.

**2.4 Emissions from incremental transportation in the year y**

As described above, $PE_{transp,y}$ is excluded, hence $PE_{transp,y}$ = 0 for the project.

**2.5 Emissions from the storage of manure**

As described above, $PE_{storage,y}$ is excluded, hence emissions from the storage of manure are not accounted for, $PE_{storage,y}$ = 0 $tCO_2e$.

So, the ex-ante estimated project emission shall calculate as per equation (12):

$PE_y = PE_{PL,y} + PE_{flare,y} + PE_{power,y} + PE_{transp,y} + PE_{storage,y} = PE_{PL,y} = 6{,}185\ tCO_2e$

**3. Calculation of leakage**

As per "Project and leakage emissions from anaerobic digesters" (version 02.0), the leakage emissions associated with the anaerobic digester depend on how the digestate is managed. The leakage emissions include emissions associated with storage of digestate and composting of the digestate.

Digestate from the anaerobic digesters of the project will be used to produce fertilizer as soon as digestate is removed from the digesters. Digestate will not be stored under anaerobic conditions. In addition, the project does not involve composting of digestate. Therefore, leakage emissions of the project associated with the anaerobic digester is not accounted for.

Thus, $LE_y$ = 0.

**4. Calculation of emission reductions (Ex ante calculation)**

Estimated annual emission reductions of the project can be calculated as following equation (21):

$$ER_y = BE_y - PE_y - LE_y = BE_y - PE_y \tag{21}$$

| Year | Estimated baseline emissions or removals ($tCO_2e$) | Estimated project emissions or removals ($tCO_2e$) | Estimated leakage emissions ($tCO_2e$) | Estimated net GHG emission reductions or removals ($tCO_2e$) |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| 21-Mar-2021~ 31-Dec-2021 | 49,682 | 4,846 | 0 | 44,835 |
| 1-Jan-2022~ 31-Dec-2022 | 63,405 | 6,185 | 0 | 57,220 |
| 1-Jan-2023~ 31-Dec-2023 | 63,405 | 6,185 | 0 | 57,220 |
| 1-Jan-2024~ 31-Dec-2024 | 63,405 | 6,185 | 0 | 57,220 |
| 1-Jan-2025~ 31-Dec-2025 | 63,405 | 6,185 | 0 | 57,220 |
| 1-Jan-2026~ 31-Dec-2026 | 63,405 | 6,185 | 0 | 57,220 |
| 1-Jan-2027~ 31-Dec-2027 | 63,405 | 6,185 | 0 | 57,220 |
| 1-Jan-2028~ 20-Mar-2028 | 13,723 | 1,339 | 0 | 12,385 |
| **Total** | 443,835 | 43,295 | 0 | 400,540 |

# 6 MONITORING

## 6.1 Data and Parameters Available at Validation

| Data / Parameter | $GWP_{CH4}$ |
|---|---|
| Data unit | $tCO_2e/tCH_4$ |
| Description | Global Warming Potential (GWP) of $CH_4$ applicable to the crediting period |
| Source of data | IPCC |
| Value applied | 28 |
| Justification of choice of data or description of measurement methods and procedures applied | Default value of 28 from IPCC Fifth Assessment Report (AR5). Shall be updated according to any future revision to VCS standard by VERRA. |
| Purpose of Data | Calculation of baseline emissions and project emissions |

| | |
|---|---|
| Comments | - |

| | |
|---|---|
| Data / Parameter | $D_{CH4}$ |
| Data unit | $kg/m^3$ |
| Description | $CH_4$ density |
| Source of data | AMS-III.D |
| Value applied | 0.67 (at 20 ºC and 1 atm pressure) |
| Justification of choice of data or description of measurement methods and procedures applied | - |
| Purpose of Data | Calculation of baseline emission and project emission from Physical leakage of biogas |
| Comments | - |

| | |
|---|---|
| Data / Parameter | $UF_b$ |
| Data unit | - |
| Description | Model correction factor to account for model uncertainties |
| Source of data | AMS-III.D |
| Value applied | 0.94 |
| Justification of choice of data or description of measurement methods and procedures applied | - |
| Purpose of Data | Calculation of baseline emissions and project emissions |
| Comments | - |

| | |
|---|---|
| Data / Parameter | $MCF_j$ |
| Data unit | - |

| | |
|---|---|
| Description | Annual methane conversion factor (MCF) for the baseline animal manure management system j |
| Source of data | 2019 refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories |
| Value applied | 76% |
| Justification of choice of data or description of measurement methods and procedures applied | No country or regional specific value is available. Default value from table 10.17 of 2019 refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 Chapter 10 is applied.<br>The annual average temperature of baseline site where anaerobic manure treatment facility is located is around 13°C[22] and ratio of potential evapotranspiration to precipitation is less than 1. The climate zone is warm temperature dry. Thus, the corresponding annual methane conversion factor (MCF) is 76%. |
| Purpose of Data | Calculation of baseline emissions and project emissions |
| Comments | - |

| | |
|---|---|
| Data / Parameter | $B_{0,LT}$ |
| Data unit | $m^3$ $CH_4$/kg-VS |
| Description | Maximum methane producing potential of the volatile solid generated for animal type LT |
| Source of data | Table 10.16 of 2019 refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 Chapter 10 |
| Value applied | <table><tr><td>Animal type</td><td>$B_{0,LT}$</td></tr><tr><td>Dairy cattle</td><td>0.13</td></tr></table> |
| Justification of choice of data or description of measurement methods and procedures applied | No country or regional specific value is available. Default values from table 10.16 of 2019 refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories volume 4 Chapter 10 are applied.<br>The project adopts Asian data. |
| Purpose of Data | Calculation of baseline emissions and project emissions |

[22] http://www.weixian.gov.cn/article/256/42972.html

| | |
|---|---|
| Comments | - |

| | |
|---|---|
| Data / Parameter | $MS\%_{BL,i}$ |
| Data unit | - |
| Description | Fraction of manure handled in baseline animal manure management system j |
| Source of data | FSR |
| Value applied | 100% |
| Justification of choice of data or description of measurement methods and procedures applied | - |
| Purpose of Data | Calculation of baseline emissions and project emissions |
| Comments | - |

| | |
|---|---|
| Data / Parameter | $N_{da,y}$ |
| Data unit | Day |
| Description | Number of days animal is alive in the farm in the year y |
| Source of data | Project proponents |
| Value applied | 365 |
| Justification of choice of data or description of measurement methods and procedures applied | - |
| Purpose of Data | To calculate the annual average number of animals ($N_{LT,y}$) |
| Comments | - |

| | |
|---|---|
| Data / Parameter | $VS_{default}$ |
| Data unit | kg/hd/day |

| | |
|---|---|
| Description | Default value for the volatile solid excretion rate per day on a dry-matter basis for a defined livestock population |
| Source of data | 2019 refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories |
| Value applied | Animal type: Dairy cattles; $VS_{default}$: 3.474 |
| Justification of choice of data or description of measurement methods and procedures applied | Calculated via equation 10.22A in 2019 Refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories.<br>$VS_{default} = VS_{rate(T)} * TAM / 1000 = 9.0 * 386 / 1000$<br>TAM: Default Values for livestock weight for Animal Categories in Asia area is 386kg(mean) for dairy cattle<br>$VS_{rate(T)}$ = Default Values for Volatile Solid Excretion Rate, 9.0 kg VS 1000 kg animal mass$^{-1}$ day$^{-1}$ for dairy cattle sourced from Volume 4, chapter 10, table 10.13A of 2019 Refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories. |
| Purpose of Data | Calculation of baseline emissions and project emissions |
| Comments | - |

| | |
|---|---|
| Data / Parameter | $EF_{grid,OM,y}$ |
| Data unit | t $CO_2$/MWh |
| Description | Operating margin $CO_2$ emission factor in year y |
| Source of data | "2019 Baseline Emission Factors for Regional Power Grids in China" published by Ministry of Ecology and Environment of China |
| Value applied | 0.9419 |
| Justification of choice of data or description of measurement methods and procedures applied | Official and authoritative statistic data. |
| Purpose of Data | Calculation of baseline emissions and project emissions |
| Comments | According to section 5.1, the ex-ante option is selected to calculate $EF_{grid, OM,y}$ and fixed in the crediting period. |

| | |
|---|---|
| Data / Parameter | $EF_{grid,BM,y}$ |

| | |
|---|---|
| Data unit | t $CO_2$/MWh |
| Description | Build margin $CO_2$ emission factor in year y |
| Source of data | "2019 Baseline Emission Factors for Regional Power Grids in China" published by Ministry of Ecology and Environment of China |
| Value applied | 0.4819 |
| Justification of choice of data or description of measurement methods and procedures applied | Official and authoritative statistic data. |
| Purpose of Data | Calculation of baseline emissions and project emissions |
| Comments | According to section 5.1, the ex-ante option is selected to calculate $EF_{grid,BM,y}$ and fixed in the crediting period. |

| | |
|---|---|
| Data / Parameter | $EF_{grid,CM,y}$ ($EF_{grid,y}$, $EF_{EL,j,y}$) |
| Data unit | t $CO_2$/MWh |
| Description | Combined margin emission factor for the grid in year y |
| Source of data | Calculated based on "Tool to calculate the emission factor for an electricity system" (version 07.0). |
| Value applied | 0.7119 |
| Justification of choice of data or description of measurement methods and procedures applied | $EF_{grid,CM,y}$ is calculated based on $EF_{grid,OM,y}$ and $EF_{grid,BM,y}$ as per the latest version of "Tool to calculate the emission factor for an electricity system" |
| Purpose of Data | Calculation of baseline emissions and project emissions |
| Comments | According to section 5.1, the ex-ante option is selected to calculate $EF_{grid,OM,y}$ and $EF_{grid,BM,y}$ and fixed in the crediting period, thus the $EF_{grid,CM,y}$ is fixed during the crediting period. |

| | |
|---|---|
| Data / Parameter | $NCV_{CH4}$ |
| Data unit | MJ/$Nm^3$ |
| Description | NCV of methane |

| | |
|---|---|
| Source of data | AMS-III.D |
| Value applied | 35.9 |
| Justification of choice of data or description of measurement methods and procedures applied | Default value from methodology is applied |
| Purpose of Data | Calculation of emission reductions |
| Comments | - |

| | |
|---|---|
| Data / Parameter | $TDL_{j,y}$ |
| Data unit | - |
| Description | Average technical transmission and distribution losses for providing electricity to source j in year y |
| Source of data | Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation (version 03.0) |
| Value applied | 20% |
| Justification of choice of data or description of measurement methods and procedures applied | Default value of 20% is applied |
| Purpose of Data | Calculation of project emissions |
| Comments | - |

| | |
|---|---|
| Data / Parameter | $EE_y$ |
| Data unit | % |
| Description | Energy Conversion Efficiency of the project equipment |
| Source of data | AMS-III.D |
| Value applied | 40% |
| Justification of choice of data or description of | Default value from methodology is applied |

| | |
|---|---|
| measurement methods and procedures applied | |
| Purpose of Data | Calculation of emission reductions |
| Comments | - |

| | |
|---|---|
| Data / Parameter | $SPEC_{flare}$ |
| Data unit | Temperature - °C<br>Flow rate or heat flux - kg/h or $m^3$ /h<br>Maintenance schedule - number of days |
| Description | Manufacturer's flare operating specifications for temperature, flow rate and maintenance schedule |
| Source of data | Flare manufacturer |
| Value applied | Temperature – 500°C ~1270°C<br>Flow rate or heat flux - 1100 $m^3$ /h<br>Maintenance schedule –every 12-24 months (around 365days – 730 days) based on operating conditions |
| Justification of choice of data or description of measurement methods and procedures applied | Default value is applied according to the methodology. |
| Purpose of Data | Calculation of project emission |
| Comments | - |

| | |
|---|---|
| Data / Parameter | $\eta_{flare,m}$ |
| Data unit | % |
| Description | Flare efficiency in minute m |
| Source of data | Default value of methodological tool 06" Project emissions from flaring (Version 04.0)" |
| Value applied | When the flame is detected in the minute m and the temperature of the flare and the flow rate of the residual gas to the flare ($F_{RG,m}$) is within the manufacturer's operating specification for the flare ($SPEC_{flare}$) in the minute m: 90%<br>Otherwise: 0% |

| | |
|---|---|
| Justification of choice of data or description of measurement methods and procedures applied | Default value from methodology is applied |
| Purpose of Data | Calculation of project emission |
| Comments | - |

| | |
|---|---|
| Data / Parameter | $f_{CH4,default}$ |
| Data unit | $m^3$ $CH_4$ / $m^3$ biogas corrected to reference conditions |
| Description | Default value for the fraction of methane in the biogas |
| Source of data | The default value was derived based on reported values from registered projects and research papers (Davidsson, 2007) |
| Value applied | 0.6 |
| Justification of choice of data or description of measurement methods and procedures applied | - |
| Purpose of Data | Calculation of project emissions |
| Comments | Use this value for Option 2 of the step “Determination of the quantity of methane produced in the digester” |

| | |
|---|---|
| Data / Parameter | $R_u$ |
| Data unit | Pa.m$^3$/kmol.K |
| Description | Universal ideal gases constant |
| Source of data | “Tool to determine the mass flow of a greenhouse gas in a gaseous stream” (Version 03.0) |
| Value applied | 8,314 |
| Justification of choice of data or description of measurement methods and procedures applied | - |
| Purpose of Data | Calculation of project emissions |
| Comments | - |

| | |
|---|---|
| Data / Parameter | $MM_i$ |
| Data unit | kg/kmol |
| Description | Molecular mass of greenhouse gas $CH_4$ |
| Source of data | "Tool to determine the mass flow of a greenhouse gas in a gaseous stream" (Version 03.0) |
| Value applied | 16.04 |
| Justification of choice of data or description of measurement methods and procedures applied | - |
| Purpose of Data | Calculation of project emissions |
| Comments | - |

| | |
|---|---|
| Data / Parameter | $P_n$ |
| Data unit | Pa |
| Description | Total pressure at normal conditions |
| Source of data | "Tool to determine the mass flow of a greenhouse gas in a gaseous stream" (Version 03.0) |
| Value applied | 101,325 Pa |
| Justification of choice of data or description of measurement methods and procedures applied | - |
| Purpose of Data | Calculation of project emissions |
| Comments | - |

| | |
|---|---|
| Data / Parameter | $T_n$ |
| Data unit | K |
| Description | Temperature at normal conditions |
| Source of data | "Tool to determine the mass flow of a greenhouse gas in a gaseous stream" (Version 03.0) |

| | |
|---|---|
| Value applied | 273.15 |
| Justification of choice of data or description of measurement methods and procedures applied | - |
| Purpose of Data | Calculation of emissions |
| Comments | - |

| | |
|---|---|
| Data / Parameter | $\rho_{i,n}$ & $\rho_{RG,n}$ |
| Data unit | Kg gas i/m³ gas I, i=RG=$CH_4$ |
| Description | Density of greenhouse gas $CH_4$ in the gaseous stream at normal conditions (0℃ and 1 atm) |
| Source of data | Calculated in the section 5 of this joint PDMR as per the latest version of "Tool to determine the mass flow of a greenhouse gas in a gaseous stream" (Version 03.0) |
| Value applied | 0.71566 |
| Justification of choice of data or description of measurement methods and procedures applied | - |
| Purpose of Data | Calculation of project emission from flaring |
| Comments | Applicable to Option A, $\rho_{i,n=}\ \rho_{RG,n} = 101{,}325\text{Pa} \times 16.04\text{kg/kmol}/(8314\text{Pa.}\,\text{m}^3/\text{kmol.}\,\text{K} \times 273.15\text{K}) = 0.71566\ \text{kg gas i/m}^3\ \text{gas i}$ |

## 6.2 Data and Parameters Monitored

The project site is at the east side of the pasture and waste from the pasture will all enter the receiving tank of the pre-treatment system through tight pipeline, thus there is no emission from transportation of animal manure between livestock farms and the project site. $PE_{transp,y}$ = 0. Thus, the related parameters of $Q_y$, $CT_y$, $DAF_w$, $Q_{res\cdot waste,y}$, $CT_{res,waste,y}$, $DAF_{res\cdot waste}$ are not been monitored.

| | |
|---|---|
| Data / Parameter | $MS\%_{i,y}$ |
| Data unit | - |

| | |
|---|---|
| Description | Fraction of manure handled in system i in year y |
| Source of data | Project proponents |
| Description of measurement methods and procedures to be applied | all manure would be handled in this project, 100% is applied for $MS\%_{i,y}$ in ex-ante estimation. This parameter will be monitored in verification period. |
| Frequency of monitoring/recording | - |
| Value applied | 100% |
| Monitoring equipment | - |
| QA/QC procedures to be applied | - |
| Purpose of data | Calculation of baseline emissions and project emissions |
| Calculation method | - |
| Comments | - |

| | |
|---|---|
| Data / Parameter | $N_{p,y}$ |
| Data unit | Number |
| Description | Number of animals produced annually of type LT for the year y |
| Source of data | Project proponents |
| Description of measurement methods and procedures to be applied | The number of Dairy cattles produced in the farm will be recorded manually by the responsible staff. |
| Frequency of monitoring/recording | Annually, based on monthly records |
| Value applied | 20,000 estimated ex ante and will be monitored ex post |
| Monitoring equipment | - |
| QA/QC procedures to be applied | - |
| Purpose of data | To calculate the annual average number of animals ($N_{LT,y}$) |
| Calculation method | - |

| | |
|---|---|
| Comments | - |

| | |
|---|---|
| Data / Parameter | $nd_y$ |
| Data unit | number |
| Description | The number of days the treatment plant was operational in year y. |
| Source of data | Project proponents |
| Description of measurement methods and procedures to be applied | 365 days used for ex-ante estimation. The actual number of days the treatment plant was operationally used in the monitoring periods will be monitored and recorded by staff. |
| Frequency of monitoring/recording | Annually, based on daily records and monthly aggregation |
| Value applied | 365 |
| Monitoring equipment | - |
| QA/QC procedures to be applied | - |
| Purpose of data | To estimate the annual volatile solid excretions for livestock LT entering all animal waste management systems on a dry matter weight basis ($VS_{LT,y}$). |
| Calculation method | - |
| Comments | - |

| | |
|---|---|
| Data / Parameter | $EG_y$ |
| Data unit | MWh |
| Description | Total electricity generated from the recovered biogas in year y |
| Source of data | Electricity meter |
| Description of measurement methods and procedures to be applied | The parameter is not monitored by the project owner, it will be replaced by the parameter $EG_{PJ,y}$. |
| Frequency of monitoring/recording | Continuously monitored and monthly recorded |

| | |
|---|---|
| Value applied | 27,000 (Ex-ante estimation) |
| Monitoring equipment | Electricity meter |
| QA/QC procedures to be applied | - |
| Purpose of data | Calculation of emission reductions |
| Calculation method | - |
| Comments | $EG_y$ is the total electricity generated from the recovered biogas in year y. $EG_{PJ,y}$ is the quantity of electricity generation supplied by the project to the grid in year y.<br>Using $EG_{PJ,y}$ to replace $EG_y$ is more conservative. In addition, no matter it is $EG_y$ or $EG_{PJ,y}$, it does not influence the amount of the final emission reduction since after the minimization *according to the* methodology of $(BE_{y,ex\,post} - PE_{y,ex\,post})$ and $(MD_y - PE_{power,y,ex\,post})$, the value of $(BE_{y,ex\,post} - PE_{y,ex\,post})$ is taken. |

| | |
|---|---|
| Data / Parameter | $EC_{PJ,j,y}$ |
| Data unit | MWh |
| Description | Quantity of electricity consumed by the project in year y |
| Source of data | Electricity meter |
| Description of measurement methods and procedures to be applied | Continuously measured by electricity meter and at least monthly recorded |
| Frequency of monitoring/recording | Continuously measured and at least monthly recorded |
| Value applied | 0 (estimated ex-ante) |
| Monitoring equipment | Electricity meter |
| QA/QC procedures to be applied | The data will be cross checked with Electricity Transaction Note. The metering instruments with accuracy of 0.2S (satisfied the requirement of no less than 0.5s) will be calibrated annually in accordance with "DL/T448–2016 Technical Administrative Code of Electric Energy Metering" which is the latest version of "Technical administrative code of electric energy metering". |
| Purpose of data | Calculation of project emissions |

| | |
|---|---|
| Calculation method | - |
| Comments | - |

| | |
|---|---|
| Data / Parameter | $EG_{PJ,y}$ |
| Data unit | MWh |
| Description | Quantity of electricity generation supplied by the project to the grid in year y |
| Source of data | Electricity meter |
| Description of measurement methods and procedures to be applied | Measured by electricity meter continuously and monthly recorded |
| Frequency of monitoring/recording | Continuous monitoring and at least monthly recording |
| Value applied | - |
| Monitoring equipment | Electricity meter |
| QA/QC procedures to be applied | The data will be cross checked with Electricity Transaction Note. The metering instruments with accuracy of 0.2S (satisfied the requirement of no less than 0.5s) will be calibrated annually in accordance with "DL/T448-2016 Technical Administrative Code of Electric Energy Metering" which is the latest version of "Technical administrative code of electric energy metering". |
| Purpose of data | Calculation of baseline emissions |
| Calculation method | - |
| Comments | - |

| | |
|---|---|
| Data / Parameter | $F_{CH4,RG,m}$ |
| Data unit | t $CH_4$ |
| Description | Mass flow of methane in the residual gas in the minute m |
| Source of data | Calculated |
| Description of measurement methods | - |

| | |
|---|---|
| and procedures to be applied | |
| Frequency of monitoring/recording | - |
| Value applied | 0 when there is no malfunction and the flare system does not operate |
| Monitoring equipment | - |
| QA/QC procedures to be applied | - |
| Purpose of data | Calculation of project emissions. |
| Calculation method | - |
| Comments | - |

| | |
|---|---|
| Data / Parameter | $V_{RG,t,db}$ |
| Data unit | $m^3$ dry gas/h |
| Description | Volumetric flow of the residue gas in time interval t on a dry basis at normal conditions |
| Source of data | Measured by a flow meter continuously and converted automatically by the recovery monitoring system |
| Description of measurement methods and procedures to be applied | The parameters of $V_{t,db,n}$ is measured by a flow meter continuously, which will be converted automatically into standard value $V_{t,db,n}$ (at the normal condition of 0℃ and 101,325 Pa) by the recovery monitoring system |
| Frequency of monitoring/recording | Monitoring continuously and recording monthly |
| Value applied | The data will be monitored ex post in the monitoring period. In the estimated net GHG emission reduction phase, the amount of biogas is estimated and calculated as per section 5, please refer to ER calculation sheet for details. |
| Monitoring equipment | Flow meter |
| QA/QC procedures to be applied | Periodic calibration against a primary device provided by an independent accredited laboratory is mandatory for all projects applying large scale methodology(ies). Calibration will be carried out once every two years according to manufacturer's specifications.<br><table><tr><td>Type</td><td>Serial number</td><td>Date of calibration</td><td>Validity</td></tr></table> |

| | | | | |
|---|---|---|---|---|
| | HGF-3000F | 200051040051 | 06-Apr-2020;05-Apr-2022 | 05-Apr-2022;04-Apr-2024 |

| | |
|---|---|
| Purpose of data | Calculation of project emissions |
| Calculation method | - |
| Comments | - |

| | |
|---|---|
| Data / Parameter | Operation hours (to calculate $F_{CH4,RG,m}$) |
| Data unit | Hours |
| Description | The total operation hours of the flare (to calculate $F_{CH4,RG,m}$) |
| Source of data | Project participants |
| Description of measurement methods and procedures to be applied | - |
| Frequency of monitoring/recording | Monitoring hourly |
| Value applied | The data will be monitored ex post in the monitoring period. |
| Monitoring equipment | - |
| QA/QC procedures to be applied | - |
| Purpose of data | Calculation of project emissions. |
| Calculation method | - |
| Comments | - |

## 6.3 Monitoring Plan

The monitoring plan presented in this PD assures that real, measurable, long-term GHG emission reductions can be monitored, recorded and reported. It is a crucial procedure to identify the final VCUs of the project. This monitoring plan will be implemented by the project owner during the project operation. The details of the monitoring plan are specified as follows:

**(A) Monitoring structure**

The Project owner organizes a specific VCS team in project development department to be responsible for data collection, supervision and witness the whole process of data measuring and recording. A VCS manager is appointed to take full responsibility for the overall monitoring of the project. The monitoring and measurement are to be carried out by designated monitoring officers. In addition, the Project developer appoints internal verifiers who is responsible for internal check of the measurement, collection of relevant receipts and invoices, and the calculation of the emission reductions. A monitoring and management manual of the project that identifies detailed duties and responsibilities of the relevant parties is developed and served as the basis of the project monitoring. Figure 6-1 shows the operation and management structure of the Project.

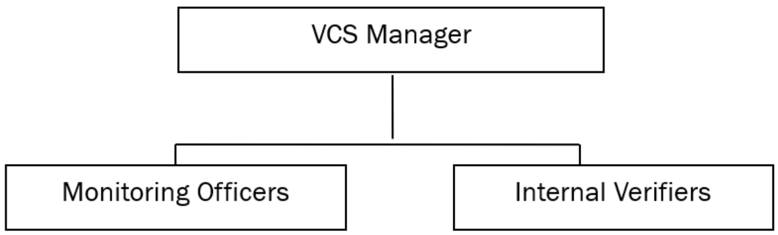


**Figure 6-1 Operation and management structure of the project**

(B) Data and parameters to be monitored

Data and parameters to be monitored are listed below. Figure 6-2 shows the positions of the monitoring instruments and Table 6-1 lists the corresponding parameters monitored:

**Table 6-1 Data and parameters to be monitored**

| Parameter to be Monitored | Description |
|---|---|
| $EG_y/EG_{PJ,y}$ | Quantity of electricity generation supplied by the project to the grid in year y is measured by electricity meter i*nstalled between the project site and the grid* continuously and w*ill be* monthly recorded<br>The metering instruments with accuracy of 0.2S (satisfied the requirement of no less than 0.5 s) will be calibrated annually in accordance with the “DL/T448–2016 Technical Administrative Code of Electric Energy Metering” which is the latest version of “Technical administrative code of electric energy metering”. |
| $EC_{PJ,j,y}$ | Quantity of electricity consumed by the project in year y is continuously measured by electricity meter i*nstalled between the project site and the grid* continuously and w*ill be* monthly recorded |

| | |
|---|---|
| | The metering instruments with accuracy of 0.2S (satisfied the requirement of no less than 0.5 s) will be calibrated annually in accordance with the “DL/T448–2016 Technical Administrative Code of Electric Energy Metering” which is the latest version of “Technical administrative code of electric energy metering”. |
| $V_{RG,t,db}$ | The flow meter will be calibrated every two years as per Verification Regulation of Thermal Mass Gas Flowmeters (JJG 1132) |

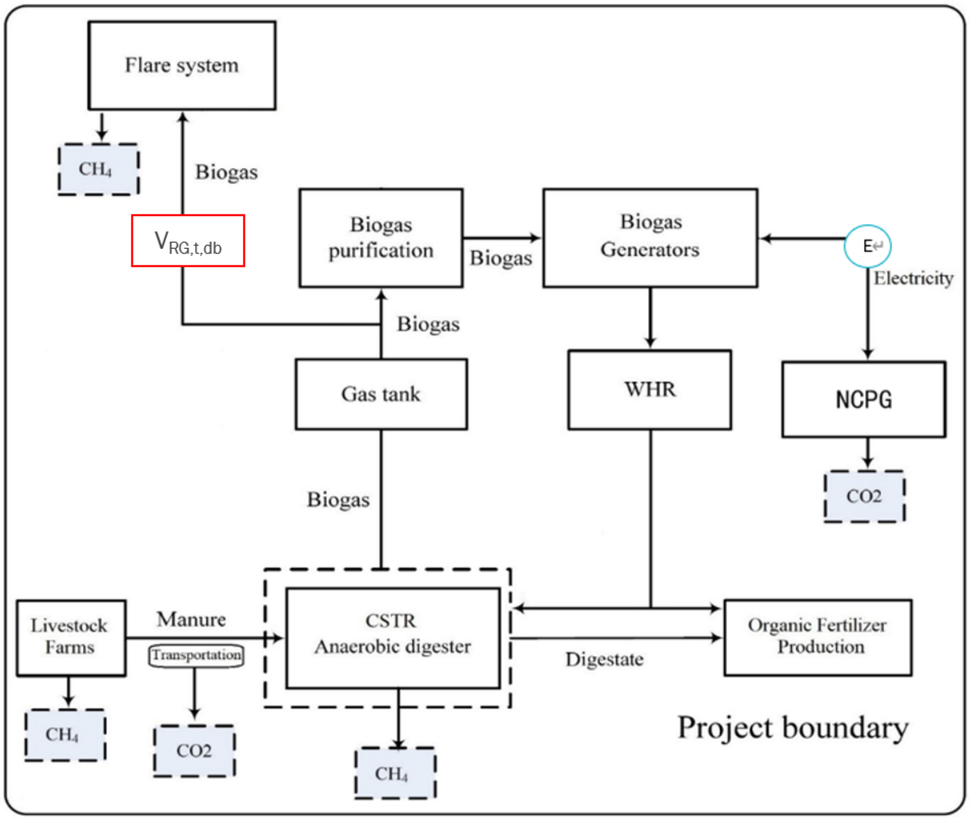


Figure 6-2 Project monitoring diagram

**(C) Data collection**

Monitoring officers are responsible for data collection. Designated teams will read and collect the monitored data regularly. The computer system will automatically monitor and record relevant meter data. Automatic records will serve as the main data source for emission reductions calculation. All data files, relevant receipts will be collected by a designated monitoring officer, who will prepare backup in time and archive all documents properly.

**(D) Quality assurance**

All metering equipment for monitoring will be chosen in accordance with VCS requirements and will be calibrated regularly for accuracy by qualified party according to the national regulations. To assist in future verifications, the Project owner will preserve the calibration records, along with the data files of project monitoring.

Error check routines will be established on site and at the point of data storage to detect data measuring/transmission failures as well as malfunctions. In the case of malfunction of the meters, the meter supplier will provide technical support to engage the problem promptly and emission reductions during the corresponding period will be calculated conservatively.

The installation of the electricity metering equipment will fulfill the requirements of "*DL/T448–2016 Technical Administrative Code of Electric Energy Metering*". The accuracy should not be less than 0.5s. All the meters will be checked and maintained periodically.

**(E) Data file management**

All monitoring data will be electronically filed by the end of each month and the electronic data files will be archived in both disk copy and printed hard copy. Other documents in paper e.g., forms and environment assessment reports will be preserved as well. All data collected as part of monitoring will be archived electronically and be kept at least for 2 years after the end of the crediting period. The Project owner will provide original records and documents if necessary.

# 7 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

## 7.1 Data and Parameters Monitored

| | |
|---|---|
| **Data / Parameter** | $MS\%_{i,y}$ |
| **Data unit** | - |
| **Description** | Fraction of manure handled in system i in year y |
| **Value applied:** | 100% |
| **Comments** | - |

| | |
|---|---|
| **Data / Parameter** | $N_{p,y}$ |
| **Data unit** | number |

| | |
|---|---|
| Description | Number of animals produced annually of type LT for the year y |
| Value applied: | Period / Value(head): 21-Mar-2021~31-Dec-2021: 12,059; 01-Jan-2022~30-Sep-2022: 12,093 |
| Comments | This parameter is recorded monthly according to daily report of cattle dynamics from Leyuan ANIMAL Husbandry Weixian Co., Ltd. NO. 4 Pasture. |

| Period | Value(head) |
|---|---|
| 21-Mar-2021~31-Dec-2021 | 12,059 |
| 01-Jan-2022~30-Sep-2022 | 12,093 |

| | |
|---|---|
| Data / Parameter | $nd_y$ |
| Data unit | number |
| Description | The number of days the treatment plant was operational in year y |
| Value applied: | |
| Comments | - |

| Period | Value(day) |
|---|---|
| 21-Mar-2021~31-Dec-2021 | 285 |
| 01-Jan-2022~30-Sep-2022 | 273 |

| | |
|---|---|
| Data / Parameter | $EG_y$ |
| Data unit | MWh |
| Description | Total electricity generated from the recovered biogas in year y |
| Value applied: | - |
| Comments | The parameter is not monitored by the project owner, it has been replaced by the parameter $EG_{PJ,y}$.<br>$EG_y$ is the total electricity generated from the recovered biogas in year y. $EG_{PJ,y}$ is the quantity of electricity generation supplied by the project to the grid in year y. |

| | |
|---|---|
| | Using $EG_{PJ,y}$ to replace $EG_y$ is more conservative. In addition, no matter it is $EG_y$ or $EG_{PJ,y}$, it does not influence the amount of the final emission reduction since after the minimization *according to the* methodology of $(BE_{y,ex\ post} - PE_{y,ex\ post})\text{and}(MD_y - PE_{power,y,ex\ post})$, the value of $(BE_{y,ex\ post} - PE_{y,ex\ post})$ is taken. |

| | |
|---|---|
| Data / Parameter | $EC_{PJ,j,y}$ |
| Data unit | MWh |
| Description | Quantity of electricity consumed by the project in year y |
| Value applied: | <table><tr><th>Period</th><th>EC<sub>PJ,j,y</sub> (MWh)</th></tr><tr><td>21-Mar-2021~31-Dec-2021</td><td>0.000</td></tr><tr><td>01-Jan-2022~30-Sep-2022</td><td>0.000</td></tr><tr><td>Total</td><td>0.000</td></tr></table> |
| Comments | During this monitoring period, the data is zero and is confirmed by the staff of the project and staff from local power grid company. The metering instruments have been calibrated annually in accordance with the latest version “Technical administrative code of electric energy metering”.<br><br>The information of electricity meter:<br><br>The electricity meter is at the project site between the project and NCPG. This parameter is monitored continuously and recorded monthly. The cut-off time of the electricity meter E is at 24:00 of the last day of each month.<br><br><table><tr><th>Type</th><th>Serial number</th><th>Accuracy class</th><th>Date of calibration</th><th>Validity</th></tr><tr><td>DSZ178</td><td>4300036005</td><td>0.2S</td><td>22-Dec-2020;20-Dec-2021</td><td>21-Dec-2021;19-Dec-2022</td></tr></table> |

| | |
|---|---|
| Data / Parameter | $EG_{PJ,y}$ |

| | |
|---|---|
| **Data unit** | MWh |
| **Description** | Quantity of electricity generation supplied by the project to the grid in year y |
| **Value applied:** | <table><tr><th>Period</th><th>$EG_{PJ,y}$(MWh)</th></tr><tr><td>21-Mar-2021~31-Dec-2021</td><td>7,739.360</td></tr><tr><td>01-Jan-2022~30-Sep-2022</td><td>9,557.480</td></tr><tr><td>Total</td><td>17,296.840</td></tr></table> |
| **Comments** | The data is cross checked with Electricity Transaction Note. The metering instruments have been calibrated annually in accordance with the latest version "Technical administrative code of electric energy metering".<br><br>The information of electricity meter:<br><br>The electricity meter is at the project site between the project and NCPG. This parameter is monitored continuously and recorded monthly. The cut-off time of the electricity meter E is at 24:00 of the last day of each month.<br><br><table><tr><th>Type</th><th>Serial number</th><th>Accuracy class</th><th>Date of calibration</th><th>Validity</th></tr><tr><td>DSZ178</td><td>4300036005</td><td>0.2S</td><td>22-Dec-2020;20-Dec-2021</td><td>21-Dec-2021;19-Dec-2022</td></tr></table> |

| | |
|---|---|
| **Data / Parameter** | $F_{CH4,RG,m}$ |
| **Data unit** | t $CH_4$ |
| **Description** | Mass flow of methane in the residual gas in the minute m |
| **Value applied:** | 0 |
| **Comments** | There is no malfunction during this verification period |

| Data / Parameter | $V_{RG,t,db}$ |
|---|---|
| Data unit | $m^3$ dry gas/h |
| Description | Volumetric flow of the residue gas in time interval t on a dry basis at normal conditions |
| Value applied: | 0 |
| Comments | There is no malfunction during this verification period |

| Type | Serial number | Date of calibration | Validity |
|---|---|---|---|
| HGF-3000F | 200051040051 | 06-Apr-2020;05-Apr-2022 | 05-Apr-2022;04-Apr-2024 |

| Data / Parameter | Operation hours (to calculate $F_{CH4,RG,m}$) |
|---|---|
| Data unit | Hours |
| Description | The total operation hours of the flare (to calculate $F_{CH4,RG,m}$) |
| Value applied: | 0 |
| Comments | There is no malfunction during this verification period |

## 7.2 Baseline Emissions

According to section 5.1, baseline emissions are calculated using equation below:

$$BE_y = BE_{CH4,y} + BE_{Elec,y}$$

**1. Baseline emissions from the manure treatment processes in year y ($BE_{CH4,y}$)**

According to AMS-III.D, the emission reductions achieved in any year due to utilization of manure are the lowest value of the following:

$$\mathrm{ER_{y,ex\,post}} = \min\left[\left(\mathrm{BE_{y,ex\,post}} - \mathrm{PE_{y,ex\,post}}\right), \left(\mathrm{MD_y} - \mathrm{PE_{power,y,ex\,post}}\right)\right]$$

$BE_{y,ex\,post}$ is calculated according to the equation below:

$$BE_{y,ex\,post} = GWP_{CH4} \times D_{CH4} \times UF_b \times \sum_{j,LT} MCFj \times B_{0,LT} \times N_{LT,y} \times VS_{LT,y} \times MS\%_{i,y}$$

The value of $BE_{y,\,ex\,post}$ monitored during this monitoring period is shown in the table 7-1

Table 7-1 The value of $BE_{y,\,ex\,post}$ during this monitoring period

| Period | 21-Mar-2021~31-Dec-2021 | 01-Jan-2022~30-Sep-2022 | Unit |
|---|---|---|---|
| $GWP_{CH4}$ | 28 | 28 | $tCO_2e/tCH_4$ |
| $D_{CH4}$ | 0.00067 | 0.00067 | $t/m^3$ |
| $UF_b$ | 0.94 | 0.94 | - |
| $MCF_j$ | 76% | 76% | |
| $B_{0,LT}$ | 0.13 | 0.13 | |
| $N_{LT,y}$ | 12,059 | 12,093 | hd |
| $VS_{LT,y}$ | 990.09 | 948.40 | kg/hd/day |
| $MS\%_{i,y}$ | 100% | 100% | |
| $BE_{y,ex\,post}$ | 20,801 | 19,982 | $tCO_2e$ |

$MD_y$ is calculated using the equation below:

$$MD_y = \frac{EG_y \times 3600}{NCV_{CH4} \times EE_y} \times D_{CH4} \times GWP_{CH4}$$

For this project, the $EG_y$ has been replaced by $EG_{PJ,y}$ and the monitored value of $EG_{PJ,y}$ is shown in the table 7-2.

Table 7-2 The monitored value of $EG_{PJ,y}$

| Period | | $EG_{PJ,y}$(MWh) | | |
|---|---|---|---|---|
| | | Meter Reading Record (A) | Electricity Transaction Note Record（B) | Conservative Data（C=MIN (A,B) ） |
| 21-Mar-2021 | 30-Apr-2021 | 701.800 | 700.320 | 700.320 |
| 01-May-2021 | 31-May-2021 | 630.400 | 629.800 | 629.800 |
| 01-Jun-2021 | 30-Jun-2021 | 721.800 | 721.040 | 721.040 |
| 01-Jul-2021 | 31-Jul-2021 | 722.200 | 721.200 | 721.200 |
| 01-Aug-2021 | 31-Aug-2021 | 681,000 | 679.800 | 679.800 |
| 01-Sep-2021 | 30-Sep-2021 | 969,300 | 968.240 | 968.240 |
| 01-Oct-2021 | 31-Oct-2021 | 1,184,400 | 1,183.360 | 1,183.360 |
| 01-Nov-2021 | 30-Nov-2021 | 1,111,500 | 1,110.440 | 1,110.440 |
| 01-Dec-2021 | 31-Dec-2021 | 1,026,000 | 1,025.160 | 1,025.160 |
| Subtotal | | | | 7,739.360 |
| 01-Jan-2022 | 31-Jan-2022 | 1,058.200 | 1,057.000 | 1,057.000 |

| | | | | |
|---|---|---|---|---|
| 01-Feb-2022 | 28-Feb-2022 | 1,052.800 | 1,051.600 | 1,051.600 |
| 01-Mar-2022 | 31-Mar-2022 | 1,206.600 | 1,204.920 | 1,204.920 |
| 01-Apr-2022 | 30-Apr-2022 | 1,097.000 | 1,094.960 | 1,094.960 |
| 01-May-2022 | 31-May-2022 | 1,070.000 | 1,068.680 | 1,068.680 |
| 01-Jun-2022 | 30-Jun-2022 | 914.700 | 912.440 | 912.440 |
| 01-Jul-2022 | 31-Jul-2022 | 985.400 | 984.200 | 984.200 |
| 01-Aug-2022 | 31-Aug-2022 | 969.600 | 967.760 | 967.760 |
| 01-Sep-2022 | 30-Sep-2022 | 1,218.200 | 1,215.920 | 1,215.920 |
| Subtotal | | | | 9,557.480 |
| Total | | | | 17,296.840 |

The value of $MD_y$ monitored during this monitoring period is shown in the table 7-3

Table 7-3 The value of $MD_y$ during this monitoring period

| Period | $EG_{PJ,y}$(MWh) | $GWP_{CH4}$($tCO_2e/tCH_4$) | $D_{CH4}$($kg/m^3$) | $NCV_{CH4}$($MJ/Nm^3$) | $EE_y$ | MDy($tCO_2e$) |
|---|---|---|---|---|---|---|
| | A | B | C | D | E | F=A*3600/(D*E)*C*B*0.001 |
| 21-Mar-2021~31-Dec-2021 | 7,739.360 | 28 | 0.67 | 35.9 | 0.4 | 36,398 |
| 01-Jan-2022~30-Sep-2022 | 9,557.480 | 28 | 0.67 | 35.9 | 0.4 | 44,949 |
| Total | 17,296.840 | / | / | / | / | 81,347 |

**2. Baseline emissions associated with electricity generation in year y ($BE_{elec,y}$)**

As per AMS-I. D, baseline emissions associated with electricity generation in year y is calculated according to equation below:

$$BE_{elec,y} = EG_{PJ,y} \times EF_{grid,y}$$

The value of $BE_{elec,y}$ monitored during this monitoring period is shown in the table 7-4

Table 7-4 The value of $BE_{elec,y}$ during this monitoring period

| Period | $EG_{PJ,y}$(MWh) | $EF_{grid,CM,y}$ ($t\ CO_2/MWh$) | $BE_{Elec,y}$ |
|---|---|---|---|
| | A | B | C=A*B |
| 21-Mar-2021~31-Dec-2021 | 7,739.360 | 0.7119 | 5,509 |
| 01-Jan-2022~30-Sep-2022 | 9,557.480 | 0.7119 | 6,803 |
| Total | 17,296.840 | / | 12,312 |

## 7.3 Project Emissions

According to section 5.2, for this project, the $PE_{y,\ ex\ post}$ is calculated using the equation below:

$$PE_y = PE_{PL,y} + PE_{EC,y}$$

**1. Emissions due to physical leakage of biogas**

$$PE_{PL,y} = 0.10 \times GWP_{CH4} \times D_{CH4} \times \sum_{i,LT} B_{0,LT} \times N_{LT,y} \times VS_{LT,y} \times MS\%_{i,y}$$

The value of $PE_{PL,y}$ monitored during this monitoring period is shown in the table 7-5

Table 7-5 The value of $PE_{PL,y}$ monitored during this monitoring period

| Period | 21-Mar-2021~31-Dec-2021 | 01-Jan-2022~30-Sep-2022 | Unit |
|---|---|---|---|
| / | 0.10 | 0.10 | / |
| $GWP_{CH4}$ | 28 | 28 | $tCO_2e/tCH_4$ |
| $D_{CH4}$ | 0.00067 | 0.00067 | $t/m^3$ |
| $B_{0,LT}$ | 0.13 | 0.13 | |
| $N_{LT,y}$ | 12,059 | 12,093 | hd |
| $VS_{LT,y}$ | 990.09 | 948.40 | kg/hd/day |
| $MS\%_{i,y}$ | 100% | 100% | |
| $PE_{PL,y}$ | 2,912 | 2,798 | $tCO_2e$ |

**2. Emissions from electricity consumption**

$$PE_{EC,y} = \sum_{j} EC_{PJ,j,y} \times EF_{EL,j,y} \times (1 + TDL_{j,y})$$

For this project, the electricity generated by the project are used for self-consumption first and then supplied to the grid. The extra electricity consumption will only happen when all the generators stop running. During this monitoring period, according to the record, no such situation happened, the extra electricity consumed by the project, $EC_{PJ,j,y}$ =0.

Thus, $PE_{power,\ y,\ ex\ post}$ = $PE_{EC,y}$ =0

**3. Emissions from flaring or combustion of the biogas stream in the year y**

There is no malfunction and the flare system has not been used during this monitoring period. $PE_{flare,y}$ = 0.

## 7.4 Leakage

As described in section 5.3, leakage for this project is 0.

## 7.5 Net GHG Emission Reductions and Removals

As per section 5, $ER_{y,\ ex\ post}$ is calculated during this monitoring period using the equation and the value is shown in table 7-6 below:

$$ER_{y,ex\ post} = \min[(BE_{y,ex\ post} - PE_{y,ex\ post}), (MD_y - PE_{power,y,ex\ post})]$$

Table 7-6 The value of $ER_{y,\ ex\ post}$ during this monitoring period

| Period | $BE_{y,ex\ post}$ | $PE_{y,\ ex\ post}$ | $BE_{y,ex\ post}$ - $PE_{y,\ ex\ post}$ | $MD_y$ | $PE_{power,y,ex\ post}$ | $MD_y$ - $PE_{power,y,ex\ post}$ | $ER_{y,ex\ post}$ |
|---|---|---|---|---|---|---|---|
| | A | B | C=A-B | D | E | F=D-E | G=min(C,F) |
| 21-Mar-2021~31-Dec-2021 | 20,801 | 2,912 | 17,889 | 36,398 | 0 | 36,398 | 17,889 |
| 01-Jan-2022~30-Sep-2022 | 19,982 | 2,798 | 17,184 | 44,949 | 0 | 44,949 | 17,184 |
| Total | | | | | | | 35,073 |

Table 7-7 The total baseline emission reductions for the project are calculated

| Period | $BE_{y,ex\ post}$ | $BE_{Elec,y}$ | BE |
|---|---|---|---|
| | A | B | C=A+B |
| 21-Mar-2021~31-Dec-2021 | 20,801 | 5,509 | 26,310 |
| 01-Jan-2022~30-Sep-2022 | 19,982 | 6,803 | 26,785 |
| Total | 40,783 | 12,312 | 53,095 |

| Year | Baseline emissions or removals ($tCO_2e$) | Project emissions or removals ($tCO_2e$) | Leakage emissions ($tCO_2e$) | Net GHG emission reductions or removals ($tCO_2e$) |
|---|---|---|---|---|
| 21-Mar-2021~31-Dec-2021 | 26,310 | 2,912 | 0 | 23,398 |
| 01-Jan-2022~30-Sep-2022 | 26,785 | 2,798 | 0 | 23,987 |
| **Total** | 53,095 | 5,710 | 0 | 47,385 |

| Ex-ante emissions | Achieved emissions | Percent difference | Justification for the difference |
|---|---|---|---|

| reductions /removals | reductions /removals | | |
|---|---|---|---|
| 21-Mar-2021~30-Sep-2022 | 47,385 | -11.12% | This monitoring period is from 21-Mar-2021 to 30-Sep-2022, with totally 559 days (286 days in year 2021 and 273 days in 2022). Based on the annual estimated emission reductions, the amount of emission reductions for this monitoring period would be 87,633 $tCO_2e$[23]. The actual emission reductions in this monitoring period (559 days) are 47,385 $tCO_2e$, which is 45.93%[24] less than the estimation. One reason is that the actual installed capacity of the project during this monitoring period is 2.33MW (2 sets of 1.165MW CMM generators have been operated to generate electricity), while the total intended installed capacity of the project is 3.83MW, the estimated emission reductions for this monitoring period was re-calculated based on the actual operation capacity. The re-calculated estimated emission reductions for this monitoring period are 53,312 $tCO_2e$ (=87,633 $tCO_2e$× 2.33MW / 3.83MW). The actual emission reductions achieved in this monitoring period is 11.12%[25] lower than the re-calculated estimated emission reductions. Considering the insufficient manure wastes supply which leads to insufficient biogas, the difference is reasonable. |

[23] 87,633 $tCO_2$ = 44,835 $tCO_2$ + 57,220 $tCO_2$ *273 days/365 days

[24]-45.93% = 47,385 $tCO_2e$ / 87,633 $tCO_2e$ -1

[25]-11.12% = 47,385 $tCO_2e$ / 53,312 $tCO_2e$ -1

# APPENDIX 1: < SUPPORTING EVIDENCE >

**Evidence for SDG 7 and SDG 13**: The electricity supplied to the grid and corresponding emission reduction in this monitoring period have been listed in below table.

| Start Date | End Date | $EG_{PJ,y}$ (MWh) | Net electricity supplied to the grid | $ER_y$ ($tCO_2e$) |
|---|---|---|---|---|
| 21-Mar-2021 | 30-Sep-2022 | 17,296.840 | 17,296.840 | 47,385 |

Please refer to section 6 of this Joint PD-MR for the evidence of the detail of the project's contribution to SDG 7 and SDG 13.

**Evidence for SDG 8:** The employee list of the project has been submitted as a separate document to VVB due to confidentiality requirement of the project proponent.